University College London, 2025-2026
Econ 0039 – Advanced Macro – Franck Portier

Problem set 1: (very) Long Run growth
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Problem I – Discussing Maddison's estimates

The text in the appendix is a quite critical review of Maddison's historical estimates of income per capita. It is nevertheless a good starting point to get an idea of the methods and difficulties in constructing historical economic data.

1 – What does represent the level of 400 dollars at 1990 international prices in the work of Maddison?

2 – What is the main critique of Clark?

3 – According to Clark, to what $400 corresponds, and why is it not realistic?

4 – Explain the use of data on the evolution of humans height made by Clark.

5 – Why is urbanization an interesting statistics?

Problem II – The three stages of economic growth

In this problem, we will work with the dataset that was put together by Angus Maddison. On this page https://www.rug.nl/ggdc/historicaldevelopment/maddison/releases/maddison-project-database-2018, download the data (Excel spreadsheet or Stata)

1 – Take income per capita for the following set of countries: France, Portugal, Mexico, China, Morocco. For each country, plot the evolution of income per capita.

2 – Compute for each county average and total growth rate of per capita income for the following periods: 1-1700, 1700-1820, 1820-1913, 1913-2008. Discuss the results.

3 – Compute for each year the coefficient of variation of per capita income. Plot the evolution with time. Discuss. Note that the coefficient of variation is defined as Standard Deviation/Mean.

4 – Repeat questions 1 and 2 for another country of your choice. Discuss.

Problem III – The Malthusian Regime

The model is in discrete time. Consider the following model of joint determination of population and income per capita. The birth rate is given by:

$$b_t = \alpha_b + \beta_b y_t, \tag{1}$$

the mortality rate by:

$$m_t = \alpha_m - \beta_m y_t, \tag{2}$$

and the production function is:

$$Y_t = \alpha_y + \beta_y P_t. \tag{3}$$

Y stands for total production (or income), P for population and y for income per capita. It is assumed that all the population works and that there is non immigration nor emigration. All coefficients of the model are positive. It is assumed that $\alpha_m > \alpha_b$ and $\alpha_m - \alpha_b - \beta_y(\beta_b + \beta_m) > 0$.

1 – Discuss equations (1) to (3). Compute marginal and average productivity of labor. Comment.

2 – Compute the steady state level of total income, per capita income and population. Show graphically how those steady state values are determined.

3 – For the rest of the exercice, we assume $\alpha_b = \beta_y = 0$, $\beta_b = \beta_m = 0.5$, $\alpha_y = 1$ and $\alpha_m = \alpha$. What are the steady state levels for this configuration of parameters?

4 – Show that the model dynamics can be summarized by a first order difference equation in P_t (of the type $P_{t+1} = f(P_t)$).

5 – Study the convergence of population to its the steady state starting from a P_0 close to 0 for the following values of α: *(i)* $0 < \alpha < 1$, *(ii)* $\alpha = 1$, *(iii)* $1 < \alpha < 2$, *(iv)* $\alpha = 2$, *(v)* $\alpha > 2$.

Problem IV – A model of unified growth in which adults consume food

We have studied in section 6 of Lecture 1 a model of unified growth. One unrealistic model assumption was that adults do not consume food, but only buy food for there children. In this problem, we remove this assumption and compute again the model equilibrium.

Adults like manufacturing goods m_t, adult food f_t and children n_t. Their utility is given by

$$u_t = m_t + \gamma \log n_t + \delta \log f_t.$$

As in the lecture, the manufacturing good is the numéra ire and one children costs one unit of food, so that the budget constraint of an adult in t is :

$$p_t n_t + p_t f_t + m_t = w_t.$$

1 – Solve the utility maximization problem of an adult and derive the demand for children and the demand for agricultural goods (food for chidden and for adults).

2 – Show that total demand for food is $\frac{(\gamma+\delta)}{p_t} L_t$.

As in the lecture, technology is for food $Y_t^A = A_t^\epsilon (L_t^A)^\alpha X^{1-\alpha}$, where X is land, assumed to be in fixed supply, and normalized to $X = 1$; L^A is agricultural labor and A is productivity on agriculture. ϵ and α are between 0 and 1. The technology for the manufacturing good is $Y_t^M = M_t^\epsilon L_t^M$. Note that we assume for simplicity that land is free disposal (land rent is zero). We finally assume Learning-By-Doing, so that $A_{t+1} - A_t = Y_t^A$ and $M_{t+1} - M_t = Y_t^M$.

3 – Write the equilibrium condition on the food market and derive from that condition an expression for the share of agricultural labor in total labor $\theta_t = \frac{L_t^A}{L_t}$. How is this expression different from the one in the lecture? Discuss.

4 – Give an expression for the wages in the two sectors.

5 – Why free movements of workers between sectors will equalize the wages? From the wage equality condition, derive an expression for p_t as a function of M_t, A_t, θ_t and L_t.

6 – Using the demand for children, derive n_t as a function of M_t, A_t, and L_t. How is this expression different from the one in the lecture? Discuss.

7 – Show that the long run level of population growth is the same than in the model of Lecture 5.

Problem V – Simulation of the model of unified growth

Take the model of the last part of section 7 of Lecture 1 with $Y_t^A = \mu A_t^\epsilon (L_t^A)^\alpha X^{1-\alpha}$ and $Y_t^M = \delta M_t^\phi L_t^M$. The model solution is $n_t = \mu \left(\frac{\gamma}{\delta}\right)^\alpha \frac{A_t^\epsilon}{M_t^{\alpha\phi}(L_t)^{1-\alpha}}$ and $\theta_t = \left(\frac{n_t L_t^{1-\alpha}}{\mu A_t^\epsilon}\right)^{\frac{1}{\alpha}}$.

Assume that parameters are $\alpha = 0.8$, $\epsilon = 0.45$, $\phi = 0.3$, $\mu = 0.5$, $\delta = 1.5$, $\gamma = 3.4$ and initial values are $A_0 = 0.8$, $M_0 = 18.15$ and $L_0 = 0.014$. It is assumed that the length of the period is 25 years (one generation), so that growth factors needs to be taken at the power 1/25 to be expressed in annual terms.

1 – Use a spreadsheet (Excel, Numbers,...) or the scientific language of your choice (Matlab, Python, Julia, C++, Fortran,...) to reproduce the figures for time evolution of g_L, g_A, g_M, θ and w.

2 – Assume now $\gamma = 4.5$. What does it means? Repeat the simulation exercise and discuss.

Problem VI – Dynamics and Stagnation in the Malthusian Epoch

This problem follows Ashraf and Galor, The American Economic Review, Vol. 101, No. 5, 2011) and proposes a model of the Malthusian regime. I suggest that you read that paper, which is in the appendix to this problem set.

Consider an overlapping-generations economy in which activity extends over infinite discrete time. In every period, the economy produces a single homogeneous good using land and labor as inputs. The supply of land is exogenous and fixed over time, whereas the evolution of labor supply is governed by households' decisions in the preceding period regarding the number of their children.

Production occurs according to a constant-returns-to-scale technology. The output produced at time t, Y_t is

$$Y_t = (AX)^{\alpha} L_t^{1-\alpha}, \qquad 0 < \alpha < 1,$$

where L, and X are, respectively, labor and land employed in production in period t, and A measures the technological level. The technological level may capture the percentage of arable land, soil quality, climate, cultivation and irrigation methods, as well as the knowledge required for engagement in agriculture (i.e., domestication of plants and animals). Thus, AX captures the effective resources used. Output per worker produced at time t is denoted y_t.

In each period t, a generation consisting of L_t identical individuals joins the workforce. Each individual has a single parent. Members of generation t live for two periods. In the first period of life (childhood), $t-1$, they are supported by their parents. In the second period of life (parenthood), t, they inelastically supply their labor (one unit), generating an income that is equal to the output per worker, y_t, which they allocate between their own consumption and that of their children. Individuals generate utility from consumption and the number of their children:

$$u^t = c_t^{1-\gamma} n_t^{\gamma} \qquad 0 < \gamma < 1,$$

where c_t is consumption and n_t, is the number of children of an individual of generation t.

Members of generation t allocate their income between their consumption, c_t and expenditure on children, ρn_t, where ρ is the cost of raising a child. Hence, the budget constraint for a member of generation t (in the second period of life) is

$$\rho n_t + c_t \leq y_t.$$

1 – Write the utility maximisation problem of an adult and derive optimal consumption c_t and fertility n_t as a function of income per capita. What is the impact of income on fertility? Comment.

2 – Consider that the initial size of the working population is $L_0 > 0$. Explain why the law of motion of adult population is $L_{t+1} = n_t L_t$. Use the result of the previous question to derive the equilibrium law of motion of adult population $L_{t+1} = \phi(L_t)$.

3 – Show graphically in the plane (L_t, L_{t+1}) that L will converge to a steady state. Compute that steady state $\overline{L}$. Draw the time path of L starting from $L_0 < \overline{L}$.

4 – Compute steady state population density $\overline{P}_d$.

5 – Compute the law of motion of output per worker $y_{t+1} = \psi(y_t)$. Compute the steady state level of y, denoted $\overline{y}$. Interpret the result.

6 – Assume that the L_0 is at its steady state level $\overline{L}$ and that in period 0 technology A permanently increases to the level δA, $\delta > 1$. Compute y_0, n_0 and L_1. Draw the time path of adult population L_t and output per worker y_t following that increase in A.

7 – Ashraf and Galor [2011] write in their article:

> *"The Malthusian theory generates the following testable predictions:*
>
> *(i) Within a country, an increase in productivity would lead in the long run to a larger population, without altering the long-run level of income per capita.*
>
> *(ii) Across countries, those characterized by superior land productivity or a superior level of technology would have, all else equal, higher population densities in the long run, but their standards of living would not reflect the degree of their technological advancement"*

Prove that in the model we have solved, those testable predictions are correct.

8 – Ashraf and Galor [2011] are proposing in their article a test of their model predictions (*Note: CE stands for "Common Era" and is a synonym for AD.*).

> *"The empirical examination of the central hypothesis of the Malthusian theory exploits exogenous sources of cross-country variation in land productivity and tech nological levels to examine their hypothesized differential effects on population density and income per capita during the time period 1-1500 CE"*

The authors first show that the Neolithic Revolution triggered a cumulative process of economic development, conferring a developmental head start to societies that experienced the agricultural transition earlier. Therefore, cross-country variation in the timing of the Neolithic Revolution can be used as a proxy for the variation in the level of technological advancement across countries during the agricultural stage of development. In simple words, the variable "Years since Neolithic transition" can be used as a measure of technology.

"Formally, the baseline specification adopted to test the Malthusian predictions regarding the effects of land productivity and the level of technology advancement on population density and income per capita are

$$\begin{aligned}\ln P_{i,t} &= \alpha_{0,t} + \alpha_{1,t}\ln T_i + \alpha_{2,t}\ln X_i + \alpha'_{3,t}\boldsymbol{\Gamma_i} + \alpha'_{4,t}\boldsymbol{D_i} + \delta_{i,t} \\ \ln y_{i,t} &= \beta_{0,t} + \beta_{1,t}\ln T_i + \beta_{2,t}\ln X_i + \beta'_{3,t}\boldsymbol{\Gamma_i} + \beta'_{4,t}\boldsymbol{D_i} + \varepsilon_{i,t}\end{aligned}$$

where $P_{i,t}$ is the population density of country i in year t; $y_{i,t}$ is country i's income per capita in year t; T_i is the number of years elapsed since the onset of agriculture in country i; X_i is a measure of land productivity for country i, based on the percentage of arable land and an index of agricultural suitability; $\boldsymbol{\Gamma_i}$ is a vector of geographical controls for country i, including absolute latitude and variables gauging access to waterways; $\boldsymbol{D_i}$ is a vector of continental dummies; and, $\delta_{i,t}$ and $\varepsilon_{i,t}$ are country-specific disturbance terms for population density and income per capita, respectively, in year t.

Table 1 below reproduces the result of the estimation of those two equations on a sample of 31 countries (I am not reporting standard errors, only significance levels). The dummy variables used are Log Absolute Latitude, Mean Distance to Nearest Coast or River, Percentage of Land within 100 km of Coast or River and the continent to which the country belongs.

Table 1: Effects on Income per Capita versus Population Density

	Dependent Variable is:					
	Log Income per Capita in:			Log Population Density in:		
	1500 CE	1000 CE	1 CE	1500 CE	1000 CE	1 CE
	(1)	(2)	(3)	(4)	(5)	(6)
Years since Neolithic transition	0.159	0.073	0.109	1.337**	0.832**	1.006**
Log Land Productivity	0.041	-0.021	-0.001	0.584***	0.364***	0.681**
Log Absolute Latitude	-0.041	0.060	-0.175	0.050	-2.140**	-2.163**
Mean Distance to Nearest Coast or River	0.215	-0.111	0.043	-0.429	-0.237	0.118
Percentage of Land within 100 km of Coast or River	0.124	-0.150	0.042	1.855**	1.326**	0.228
Continent Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	31	26	29	31	26	29
R-squared	0.66	0.68	0.33	0.88	0.95	0.89

*Notes: (i) log land productivity is the first principal component of the log of the percentage of arable land and the log of an agricultural suitability index; (ii) a single continent dummy is used to represent the Americas, which is natural given the historical period examined; (iii) regressions (2)-(3) and (5)-(6) do not employ the Oceania dummy due to a single observation for this continent in the corresponding regression samples, restricted by the availability of income per capita data; (iv) *** denotes statistical significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level, all for two-sided hypothesis tests. The absence of stars means that the coefficient is not significant at a 10 percent level.*

Interpret and comment those results through the lens of the model.

Appendix

1 − Review of Angus Maddison book *"Contours of the World Economy, 1-2030 AD: Essays in Macroeconomic History"* by Gregory Clark in Journal of Economic History, 2009

2 − "*Dynamics and Stagnation in the Malthusian Epoch*", Ashraf and Galor, The American Economic Review, Vol. 101, No. 5, 2011

Book Reviews

GENERAL AND MISCELLANEOUS

Contours of the World Economy, 1–2030 AD: Essays in Macro-Economic History. By Angus Maddison. Oxford: Oxford University Press, 2007. Pp. xii, 418. $99.00, cloth; $42.95, paper.

Angus Maddison has labored fifty years to provide ever more comprehensive and distant estimates of economic growth. His earliest estimates of GDP began in 1870, the era of modern statistical offices and censuses, and mainly concerned the better documented economies of Europe and North America.[1] But as his fame has grown, Maddison has become ever bolder in the mysteries of his craft. His statistical empire now spans every corner of the globe, and extends over millennia. *Contours of the World Economy* projects a revised set of these estimates, for every country on earth, back to 1 AD.[2]

Maddison's numbers have found wide currency with economists analyzing long-run growth. Google Scholar records over 12,000 references to his works. Even the most speculative estimates have been employed in high-profile publications, as is discussed below. *Contours of the World Economy* is a magisterial summary of this life project, and a discussion of elements of the derivation of the earlier data. It gives detail down to the level of estimates of population and GDP per capita in Belgium in 1 AD. It has, for example, graphs such as one comparing U.K. and Indonesian GDP per capita, by year, 1500–2003 (p. 132). It reports that the poorest people who ever lived were the Nepalese, whose GDP per capita in 1820 and 1870 is estimated at only $397 (in 1990 $) (p. 176). It is also, as the subtitle "essays" warns, a somewhat disjointed explanation of the patterns of history he observes, which includes such intricacies as a discussion of relative income in the provinces of the Roman Empire in 14 AD, and a discussion of the impact of Islam on African growth. The book also finds space for a history of the art of macro-measurement, beginning with the seventeenth-century English "Political Arithmeticians."

The Maddison numbers suggest that the transition to modern growth had two phases. Before 1000 AD all societies were close to a subsistence minimum GDP per capita, which he takes as $400 (1990 international prices), and income stagnated for eons.[3] Then between 1000 and 1820 there is a period of slow but persistent economic growth in Western Europe, which tripled average real incomes there by 1820. Europe in these years was gaining decisive advantage over Asia in living standards. After 1820 there was the marked acceleration of growth rates associated with the Industrial Revolution, and its spread to the rest of the world. This discovery of a two-phase process of growth has important implications for growth theorists trying to model long-run growth.

There is, however, a problem at the core of the book, and indeed at the core at the whole Maddison project for at least the last ten years. All the numbers Maddison estimates for the years before 1820 are fictions, as real as the relics peddled around Europe in the Middle Ages. Many of the numbers for the years 1820, 1870, and 1913

[1] Maddison, *Economic Growth*.

[2] It actually revised the estimates provided earlier by Maddison, *World Economy: A Millenial Perspective.*

[3] Except, as noted, curiously Nepal, with its $397 GDP per capita.

are equally fictive. Just as in the Middle Ages, there was a ready market for holy relics to lend prestige to the cathedrals and shrines of Europe—Charlemagne secured for the cathedral in Aachen, his capital, the cloak of the Blessed Virgin, and the swaddling cloths of the infant Jesus—so among modern economists there is a hunger by the credulous for numbers, any numbers however dubious their provenance, to lend support to the model of the moment. Maddison supplies that market.

How exactly Maddison created his numbers for many countries and epochs is never precisely clear. One crucial element is his assumption that the basic subsistence GDP per capita of all societies is $400 (1990 international prices). This is the fundamental constant in Maddison's world, the basic unit of human existence. Any society without a sophisticated production technology, without significant urbanization, and without a substantial rich class, or just where nothing is known, is assigned this minimum. Thus around 1000 AD the various parts of the world are mostly assumed to have incomes either of $400 (uncivilized) or $450 (civilized). Of 27 quotes of income per capita for 1000 AD for individual countries or regions, 26 lie between $400 and $450 (p. 382). Those where our ignorance is largely complete all get assigned this $400: United Kingdom, 1 AD, $400, United Kingdom, 1000 AD, $400, and Mexico, 1 AD, $400.

Why $400 is the assumed subsistence income is not explained. Maddison has no estimates from 1820 on, where data does exist, for income per person for the types of societies this number is supposed to apply to. The only societies reported to have close to these income levels in 1820 are those of Africa (and Nepal). But they have such incomes not because Maddison has data on their GDP. There is no such data for 1820. It is because in the absence of such data, he assumed that they lay close to his subsistence primitive.

Yet this subsistence assumption is vital to his whole account of the development of the world economy before 1820. Since by 1820, when we get closer to real data, almost all societies are found to have incomes well in excess of this, inevitably we have economic growth between 1000 and 1820. Had Maddison assumed subsistence was $700, there would be no growth from 1 AD to 1820.

What is that subsistence income in real terms? In 1990 US $ prices, a pound of white bread cost $0.70. So Maddison's $400 is the equivalent of 1.6 lbs of wheaten bread per person per day, or 1,500 kcal.[4] That is an extraordinarily low income, rarely observed in practice. Since most societies have inequality, the poorest in such a subsistence economy would have lived on the equivalent of much less than that daily 1.6 lbs of bread. So if the poorest people spent nothing on clothing, heat, shelter, light, and consumed only the cheapest form of calories such as bread, they would still be engaging in hard physical labor on a diet well below 1,500 kcal in the Maddison vision of subsistence.

There is ample evidence, however, that incomes even of the most "primitive" societies greatly exceeded the Maddison assumption. Modern anthropologists, for example, have recorded the daily food consumption of surviving hunter-gatherer and shifting cultivation groups. The median consumption per person was 2,340 kcal per day, well above Maddison's assumed subsistence.[5] Many of these calories came as much more expensive meat. So just measuring the value of their food consumption, hunter-gatherers, the most primitive of the primitives, lived at an income equivalent to at least double Maddison's $400.

[4] This is confirmed from historical data. Maddison estimates a U.K. GDP per capita of $1,706 in 1820. U.K. incomes then supplied the inhabitants with the daily equivalent of 6.4 lbs of wheaten bread, implying 1 lb of bread was equivalent to $0.73.

[5] Clark, *Farewell to Alms*, table 3.6, p. 50.

Human heights supply further evidence on early living standards. Heights increase with income, which increases the quantity and quality of foods. Thus the average English male around 1820, when income per capita on Maddison's measure would be $1,900, was only about 168 cm tall (66 inches), compared to 70 inches for the richest modern societies. In contrast for Indians around 1820, where Maddison reports an income of $533, average male heights were only 162 cm (64 inches).[6] What were the heights then of people supposedly living on $400 a day, who should be smaller even than the Indians and Chinese in the nineteenth century? For modern hunter-gatherers and shifting cultivators, the median is 165 cm. For Mesolithic and Neolithic Europeans, as evidenced by skeletons, it was 169 cm, taller even than the rich English of 1820. For a variety of societies observed for 1000 AD and before, when in Maddison's vision all societies hewed close to the starvation minimum, the median was 166 cm, little less than the prosperous English of 1820.[7]

For the years 1250–1820 there is also ample evidence of real wages across a variety of countries. These wages have been collected in recent years by a whole variety of economic historians: Robert Allen, Jean-Pascal Bassino, Giovanni Federico, Debin Ma, Paolo Malanima, Sevket Pamuk, and Jan Luiten van Zanden. There are also scattered sources on wages in various early localities. Wage payments are typically 50–75 percent of total income in societies. Thus these wage rates can be used to set a lower bound on real GDP in earlier societies.

The unskilled wage of preindustrial workers before 1800 is generally far above Maddisonian subsistence. Assume 300 days of work per year, 40 percent of the population working, all wages at the unskilled level, and the wage share in national income as high as 70 percent. Then a society with a GDP per capita of $400 would have an unskilled day wage of 3.4 lbs of bread. In contrast, the day wages of farm laborers in England in the 1440s were the equivalent of 20 lbs of bread per day, about six times Maddison's subsistence. For the earliest year we have evidence for England, 1209, the implied unskilled day wage was still the equivalent of 15 lbs of bread.[8] There are only a few societies that ever report real unskilled wages possibly consistent with Maddison's subsistence assumption.

Maddison, conscious of the difficulty of reconciling his assumptions about economic growth between 1 AD and 1820 with this copious preindustrial wage information, simply rejects it all as "primitive" and "almost universally rejected as a proxy for GDP per capita."[9] Instead he prefers to feel his way, ad hoc, between his GDP estimates for each society from whenever there is actual output data, and the time in the past when GDP was $400, using estimates where they exist of urbanization, or the labor share in agriculture. Thus for Britain, for example, Maddison just assumes that that the growth rate of GDP per person in 1500 to 1700 was the same as estimated by Nick Crafts and Knick Harley for 1700–1801, which interval of course includes part of the Industrial Revolution.[10] For other European countries, Maddison imposes ad hoc growth rates between 1500 and 1820. Austria, Denmark, Finland, Sweden, and Switzerland have exactly the same rate of growth of GDP per person from 1500 to 1820, 0.170 percent precisely.

For Italy, Maddison lists a GDP per capita of $1,100 for 1500, 1600, 1700, and $1,117 in 1820. Maddison presumably preferred to believe that Italian GDP per capita did not decline between the Renaissance and 1820 because Italian urbanization

[6] Ibid., table 3.8, p. 57.
[7] Ibid., tables 3.9, 3.10, pp. 59–61.
[8] Clark, "Farm Wages."
[9] Maddison, *World Economy: Historical Statistics*, p. 253.
[10] Maddison, *World Economy: A Millennial Perspective,* p. 246.

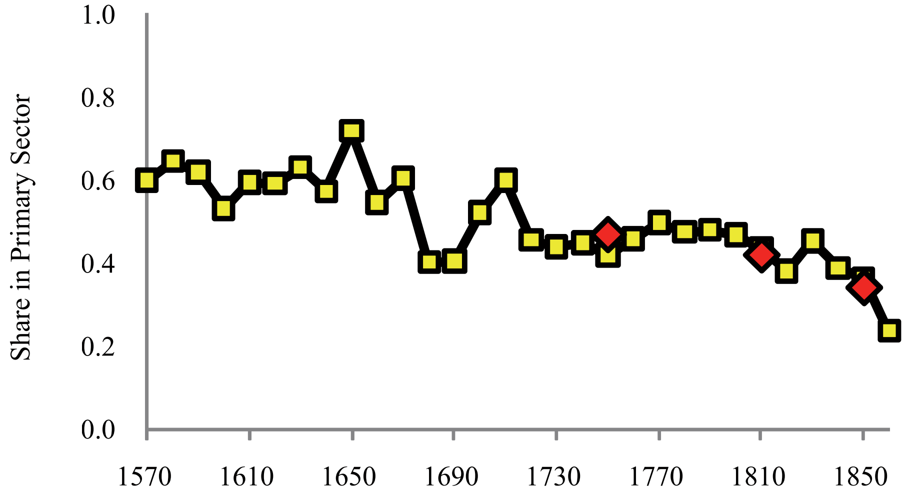

FIGURE 1
THE SHARE OF PRIMARY SECTOR EMPLOYMENT IN ENGLAND FROM WILLS, 1570–1869

Notes: The diamonds show the estimates of Shaw-Taylor and Wrigley. The wide swings in shares 1650–1739 are caused by small sample sizes in these years.

rates changed little over this interval, being around 14 percent throughout.[11] Federico and Malanima suggest, however, that real wages in north and central Italy fell by nearly 50 percent between 1500 and 1800.[12]

Urbanization is used as an indicator of per capita GDP since it is presumed to be a measure of the share of the population employed outside the primary sector. For example, for England in the years 1550 to 1800 there is a well-attested rise in the urbanization rate from 3.5 percent to 20.3 percent of the population, with in the same period no gain in rural or urban real wages.[13] The presumption from the low urbanization rate in 1550 is that the share of the population employed in the primary sector must be 70–80 percent, with a consequently low implied GDP per capita. However, it is possible to estimate the share of people employed in the primary sector for England from 1570–1860 using men's wills, which often state the testator's occupation. Figure 1 shows the implied share in primary production by decade 1570s–1860s. Also shown are recent extensive estimates by Leigh Shaw-Taylor and Anthony Wrigley for 1755, 1817, and 1851. In the decades of overlap, the estimates coincide well. But the wills show that England back in 1570, with a 3.5 percent urban share, had only 60 percent of men employed in the primary sector, compared to 46 percent by 1800. The vast majority of those in secondary and tertiary occupations were located in the countryside. Urbanization in the preindustrial world consequently is not the reliable predictor of consumption and employment patterns, and hence of income, that Maddison presumes.

England 1209–1800 is probably the best documented of all preindustrial economies. Yet we see above that even in England there is still debate about how much, if any, economic growth there was between 1209 and 1800. Those who believe there

[11] Share in towns of more than 10,000 people.
[12] Clark, *Farewell to Alms*, figure 3.3, p. 47.
[13] de Vries, *European Urbanization*, p. 39.

was significant growth in the years 1300–1800 in England have been forced to reconcile this with the contrary evidence of high early real wages by positing an "Industrious Revolution," for which there is minimal direct evidence, which dramatically increased work hours per person.[14] If the path of GDP per person even in England between 1300 and 1800 is a matter of ongoing dispute, no consensus is possible on what it was in Finland, China, India, Africa, Nepal, or anywhere else. Maddison also presents detailed population estimates, and estimates of total GDP, for the years before 1820. The population estimates are largely drawn from Colin McElvedy and Richard Jones's *Atlas of World Population History*. Inspection of that source reveals that it is similarly largely a work of imagination in these earlier years. This book, for example, happily quotes estimates of the population of the Indian subcontinent for the years 1820 and earlier. Yet examination of the sources reveals there are essentially no data on population for India, at present, for the years 1820 and before. All this material is suitable for use only in the most general and qualified way.

Despite this these numbers on GDP per capita, by country, have been widely used to lend support to theoretical models. Maddison's data for the years 1820 and earlier has been cited as tests of the theories expressed in papers published in the *American Economic Review, Quarterly Journal of Economics, Journal of Monetary Economics, Journal of Economic Growth,* and this JOURNAL, among others.[15] Somehow the Maddison numbers have an imprimatur that is completely out of line with their dubious provenance. The authors using them are happily employing as "data" what is not data at all, but the expression of one man's unexamined theory of what must have happened in the last two millennia of human history.

The interpretive essays in *Contours of the World Economy* cover a vast range of history, but mainly consist of summaries of the economic history of various parts of the world, in the light of the new GDP estimates, without any particular theme or underlying model. Thus they are not a noteworthy attraction of the book. Maddison's latest set of numbers, laid out in the tables of this book, are his claim to fame. For the reasons given above, however, any economist with enough street savvy to resist fabulous riches offered by unknown Nigerians over the internet will equally want to steer clear of these estimates.

[14] de Vries, *Industrious Revolution*; and Broadberry et al., "British Economic Growth."

[15] See, for example, Acemoglu, Johnson, and Robinson, "Rise of Europe" and "Reversal of Fortune"; Boucekkine, de la Croix, and Licandro, "Early Mortality Declines"; Falkinger and Grossman, "Institutions and Development"; Mokyr, "Long-Term Economic Growth"; Ngai, "Barriers"; and Sylla, "Financial Systems."

GREGORY CLARK, *University of California, Davis*

REFERENCES

Acemoglu, Daron, Simon Johnson, and James Robinson. "Reversal of Fortune: Geography and Institutions in the Making of the Modern World Income Distribution." *Quarterly Journal of Economics* 117, no. 4 (2002): 1231–94.

______. "The Rise of Europe: Atlantic Trade, Institutional Change, and Economic Growth." *American Economic Review* 95, no. 3 (2005): 546–79.

Allen, Robert C. "The Great Divergence in European Wages and Prices from the Middle Ages to the First World War." *Explorations in Economic History* 38, no. 4 (2001): 411–48.

Bassino, Jean-Pascal, and Debin Ma. "Japanese Wages in International Perspective, 1741–1913." *Research in Economic History* 23 (2005): 229–48.

Boucekkine, Raouf, David de la Croix, and Omar Licandro. "Early Mortality Declines at the Dawn of Modern Growth." *Scandinavian Journal of Economics* 105, no. 3 (2003): 401–18.

Broadberry, Stephen et al. "British Economic Growth, 1300–1850: Some Preliminary Estimates." Working Paper, University of Warwick, Department of Economics, 2009.

Clark, Gregory. "The Condition of the Working Class in England, 1209–2004." *Journal of Political Economy* 113, no. 6 (2005): 1307–40.

______. *A Farewell to Alms: A Brief Economic History of the World*. Princeton, NJ: Princeton University Press, 2007.

______. "Farm Wages, Population, and Economic Growth, England, 1209–1869." *Economic History Review* 60, no. 1 (2007): 97–136.

de Vries, Jan. *European Urbanization, 1500–1800*. London: Methuen, 1984.

______. *The Industrious Revolution: Consumer Behavior and the Household Economy, 1650 to the Present.* Cambridge: Cambridge University Press, 2008.

Falkinger, Josef, and Volker Grossman. "Institutions and Development: The Interaction Between Trade Regime and Political System." *Journal of Economic Growth* 10, no. 3 (2005): 231–72.

Federico, Giovanni, and Paolo Malanima. "Progress, Decline, Growth: Product and Productivity in Italian Agriculture, 1000–2000." *Economic History Review* 57, no. 3 (2004): 437–64.

Maddison, Angus. *Economic Growth in the West*. London: Allen and Unwin, 1964.

______. *The World Economy: A Millenial Perspective.* Paris: OECD, 2001.

______. *The World Economy: Historical Statistics*. Paris: OECD, 2007.

______. *Contours of the World Economy, 1–2030 AD: Essays in Macro-Economic History.* Oxford: Oxford University Press, 2008.

McEvedy, Colin, and Richard Jones. *Atlas of World Population History*. London: A. Lane, 1978.

Mokyr, Joel. "Long-Term Economic Growth and the History of Technology." In *Handbook of Economic Growth*, edited by Philippe Aghion and Steven Durlauf. Amsterdam: Elsevier, 2005.

Ngai, L. Rachel. "Barriers and the Transition to Modern Growth." *Journal of Monetary Economics* 51, no. 7 (2004): 1353–83.

Pamuk, Sevket. "Urban Real Wages Around the Eastern Mediterranean in Comparative Perspective, 1100–2000." *Research in Economic History* 23 (2005): 213–32.

Shaw-Taylor, Leigh, and E. A. Wrigley. "The Occupational Structure of England c.1750–1871: A Preliminary Report." Cambridge, England: Cambridge Group for the History of Population and Social Structure, 2008.

Sylla, Richard. "Financial Systems and Economic Modernization." This JOURNAL 62, no. 2 (2002): 277–92.

Van Zanden, Jan Luiten. "Wages and the Standard of Living in Europe, 1500–1800." *European Review of Economic History* 2 (1999):175–97.

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http://www.aeaweb.org/articles.php?doi=10.1257/aer.101.5.2003

Dynamics and Stagnation in the Malthusian Epoch†

By Quamrul Ashraf and Oded Galor*

This paper examines the central hypothesis of the influential Malthusian theory, according to which improvements in the technological environment during the preindustrial era had generated only temporary gains in income per capita, eventually leading to a larger, but not significantly richer, population. Exploiting exogenous sources of cross-country variations in land productivity and the level of technological advancement, the analysis demonstrates that, in accordance with the theory, technological superiority and higher land productivity had significant positive effects on population density but insignificant effects on the standard of living, during the time period 1–1500 CE. (*JEL* N10, N30, N50, O10, O40, O50)

The transition from an epoch of stagnation to an era of sustained economic growth has marked the onset of one of the most remarkable transformations in the course of human history. While living standards in the world economy stagnated during the millennia preceding the Industrial Revolution, income per capita has encountered an unprecedented tenfold increase in the past two centuries, profoundly altering the level and the distribution of education, health, and wealth across the globe.[1]

The Malthusian theory has been a central pillar in the interpretation of the process of development during the preindustrial era and in the exploration of the forces that brought about the transition from stagnation to growth. Nevertheless, the underlying premise of the theory, that technological progress and resource expansion during this epoch had contributed primarily to the size of the population leaving income per capita relatively unaffected in the long run, has not been tested.[2]

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[1] The transition from stagnation to growth has been examined by Galor and David N. Weil (1999, 2000), Galor and Omer Moav (2002), Gary D. Hansen and Edward C. Prescott (2002), Robert E. Lucas, Jr. (2002), Nils-Petter Lagerlöf (2003, 2006), Matthias Doepke (2004), Galor (2005), Kevin H. O'Rourke, Ahmed S. Rahman, and Alan M. Taylor (2008), Holger Strulik and Jacob Weisdorf (2008), and others, while the associated phenomenon of the Great Divergence in income per capita has been analyzed by Galor and Andrew Mountford (2006, 2008), Nico Voigtländer and Hans-Joachim Voth (2006, 2009), Ashraf and Galor (2007), and Galor (2010), among others.

[2] Recent country-specific studies provide evidence in support of *one* of the elements of the Malthusian hypothesis—the positive effect of income on fertility and its negative effect on mortality. See Nicholas Crafts and Terence C. Mills (2009) for England in the sixteenth to eighteenth centuries, Morgan Kelly and Cormac Ó Gráda (2010) in the context of medieval and early modern England, and Lagerlöf (2009) for Sweden in the eighteenth to nineteenth centuries.

The Malthusian theory, inspired by Thomas R. Malthus (1798), suggests that the worldwide stagnation in income per capita during the preindustrial epoch reflected the counterbalancing effect of population growth on the expansion of resources, in an environment characterized by the positive effect of the standard of living on population growth along with diminishing labor productivity. Periods marked by the absence of changes in the level of technology or in the availability of land were characterized by a stable population size as well as a constant income per capita, whereas periods characterized by improvements in the technological environment or in the availability of land generated only temporary gains in income per capita, eventually leading to a larger but not richer population. Technologically superior economies ultimately had denser populations but their standard of living did not reflect their technological advancement.

This research conducts a cross-country empirical analysis of the predictions of the influential Malthusian theory.[3] It exploits exogenous sources of cross-country variation in land productivity and technological levels to examine their hypothesized *differential* effects on population density versus income per capita during the time period 1–1500 CE.

In light of the potential endogeneity of population and technological progress (Boserup 1965), this research develops a novel identification strategy to examine the hypothesized effects of technological advancement on population density and income per capita. It establishes that the onset of the Neolithic Revolution that marked the transition of societies from hunting and gathering to agriculture, as early as 10,000 years ago, triggered a sequence of technological advancements that had a significant effect on the level of technology in the Middle Ages. As argued by Jared Diamond (1997), an earlier onset of the Neolithic Revolution has been associated with a developmental head start that enabled the rise of a non-food-producing class whose members were essential for the advancement of written language, science and technology, and for the formation of cities, technology-based military powers, and nation states. Thus, variations in favorable biogeographical factors (i.e., prehistoric domesticable species of wild plants and animals) that led to an earlier onset of the Neolithic Revolution across the globe are exploited as exogenous sources of variation in the onset of the Neolithic Revolution and, consequently, in the level of technological advancement during the time period 1–1500 CE.

Consistent with Malthusian predictions, the analysis uncovers statistically significant positive effects of land productivity and the technological level on population density in the years 1 CE, 1000 CE, and 1500 CE. In contrast, the effects of land productivity and technology on income per capita in these periods are not significantly different from zero. Moreover, the estimated elasticities of income per capita

[3] In contrast to the current study, which tests the Malthusian prediction regarding the positive effect of the technological environment on population density but its neutrality for income per capita, Michael Kremer (1993) examines the prediction of a Malthusian-Boserupian interaction. Accordingly, if population size has a positive effect on the rate of technological progress, as argued by Ester Boserup (1965), this effect should manifest itself as a proportional effect on the rate of population growth, *taking as given* the positive Malthusian feedback from technology to population size. Based on this premise, Kremer's study defends the role of scale effects in endogenous growth models by empirically demonstrating that the rate of population growth in the world has indeed been proportional to the level of world population throughout human history. Thus, Kremer does not test the absence of a long-run effect of the technological environment on income per capita, nor does he examine the positive effect of technology on population size.

with respect to these two channels are about an order of magnitude smaller than the corresponding elasticities of population density.

Importantly, the qualitative results remain robust to controls for the confounding effects of a large number of geographical factors, including absolute latitude, access to waterways, distance to the technological frontier, and the share of land in tropical versus temperate climatic zones, which may have had an impact on aggregate productivity either directly, by affecting the productivity of land, or indirectly via the prevalence of trade and the diffusion of technologies. Furthermore, the results are qualitatively unaffected when a direct measure of technological sophistication, rather than the timing of the Neolithic Revolution, is employed as an indicator of the level of aggregate productivity. Finally, the study establishes that the results are not driven by unobserved time-invariant country fixed effects. In particular, it demonstrates that, while the change in the level of technology between 1000 BCE and 1 CE was indeed associated with a significant change in population density over the 1–1000 CE time horizon, the level of income per capita during this time period was relatively unaffected, as suggested by the Malthusian theory.

I. The Malthusian Model

A. *The Basic Structure of the Model*

Consider an overlapping-generations economy in which activity extends over infinite discrete time. In every period, the economy produces a single homogeneous good using land and labor as inputs. The supply of land is exogenous and fixed over time, whereas the evolution of labor supply is governed by households' decisions in the preceding period regarding the number of their children.

Production.—Production occurs according to a constant-returns-to-scale technology. The output produced at time t, Y_t, is

$$Y_t = (AX)^{\alpha} L_t^{1-\alpha}; \qquad \alpha \in (0,1), \tag{1}$$

where L_t and X are, respectively, labor and land employed in production in period t, and A measures the technological level.[4] The technological level may capture the percentage of arable land, soil quality, climate, cultivation and irrigation methods, as well as the knowledge required for engagement in agriculture (i.e., domestication of plants and animals). Thus, AX captures the effective resources used in production.

Output per worker produced at time t, $y_t \equiv Y_t/L_t$, is therefore

$$y_t = (AX/L_t)^{\alpha}. \tag{2}$$

Preferences and Budget Constraints.—In each period t, a generation consisting of L_t identical individuals joins the workforce. Each individual has a single parent.

[4] The pace of technological progress, and thus the level of technology, may be determined by the size of the population (e.g., Kremer 1993; Galor and Weil 2000; Shekhar Aiyar, Carl-Johan Dalgaard, and Moav 2008) without disrupting the long-run Malthusian equilibrium.

Members of generation t live for two periods. In the first period of life (childhood), $t - 1$, they are supported by their parents. In the second period of life (parenthood), t, they inelastically supply their labor, generating an income that is equal to the output per worker, y_t, which they allocate between their own consumption and that of their children.

Individuals generate utility from consumption and the number of their (surviving) children:[5]

$$u^t = (c_t)^{1-\gamma}(n_t)^{\gamma}; \qquad \gamma \in (0,1), \tag{3}$$

where c_t is consumption and n_t is the number of children of an individual of generation t.

Members of generation t allocate their income between their consumption, c_t, and expenditure on children, ρn_t, where ρ is the cost of raising a child.[6] Hence, the budget constraint for a member of generation t (in the second period of life) is

$$\rho n_t + c_t \leq y_t. \tag{4}$$

Optimization.—Members of generation t allocate their income optimally between consumption and child rearing, so as to maximize their intertemporal utility function (3) subject to the budget constraint (4). Hence, individuals devote a fraction $(1 - \gamma)$ to consumption and a fraction γ of their income to child rearing:

$$\begin{aligned} c_t &= (1 - \gamma)y_t; \\ n_t &= \gamma y_t/\rho. \end{aligned} \tag{5}$$

Thus, in accordance with the Malthusian paradigm, income has a positive effect on the number of surviving children.

B. *The Evolution of the Economy*

Population Dynamics.—The evolution of the working population is determined by the initial size of the working population, $L_0 > 0$, and the number of (surviving) children per adult, n_t. Specifically, the size of the working population in period $t + 1$, L_{t+1}, is

$$L_{t+1} = n_t L_t, \tag{6}$$

[5] For simplicity, parents derive utility from the expected number of surviving offspring and the parental cost of child rearing is associated only with surviving children. The incorporation of parental cost for nonsurviving children would not affect the qualitative predictions of the model.

[6] If the cost of children is a time cost, then the qualitative results will be maintained as long as individuals are subjected to a subsistence consumption constraint (Galor and Weil 2000), possibly reflecting the Malthusian effects on body size (Dalgaard and Strulik 2010). If both time and goods are required to produce children, the results of the model will not be affected qualitatively. As the economy develops and wages increase, the time cost will rise proportionately with the increase in income, but the cost in terms of goods will decline. Hence, individuals will be able to afford more children.

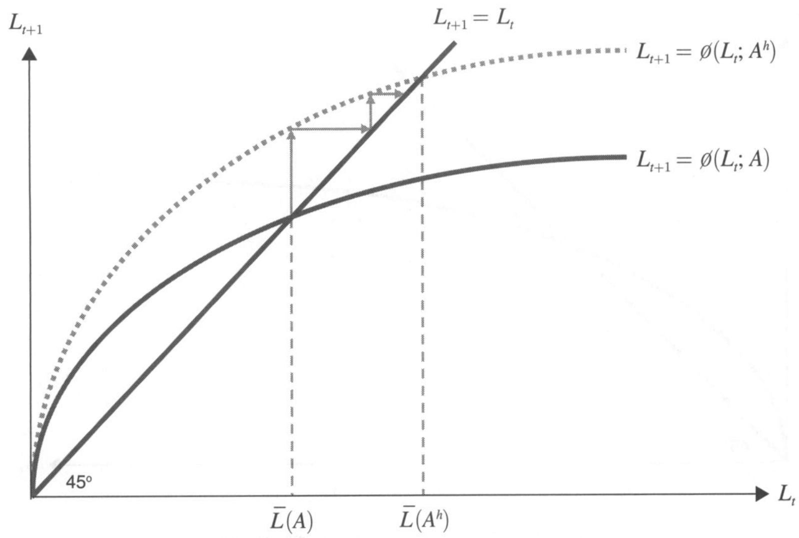

Figure 1. The Evolution of Population Size

where L_t is the size of the working population in period t, and $L_0 > 0$ is given.

Substituting (2) and (5) into (6), the time path of the working population is governed by the first-order difference equation

$$(7) \qquad L_{t+1} = (\gamma/\rho)(AX)^{\alpha}L_t^{1-\alpha} \equiv \phi(L_t;A),$$

where, as depicted in Figure 1, $\phi_L(L_t;A) > 0$ and $\phi_{LL}(L_t;A) < 0$, so $\phi(L_t;A)$ is strictly concave in L_t, and $\phi(0;A) = 0$, $\lim_{L_t \to 0} \phi_L(L_t;A) = \infty$, and $\lim_{L_t \to \infty} \phi_L(L_t;A) = 0$.

Hence, for a given level of technology, A, noting that $L_0 > 0$, there exists a unique, stable steady-state level of the adult population, $\bar{L}$:[7]

$$(8) \qquad \bar{L} = (\gamma/\rho)^{1/\alpha}(AX) \equiv \bar{L}(A),$$

and population density, $\bar{P}_d$:

$$(9) \qquad \bar{P}_d \equiv \bar{L}/X = (\gamma/\rho)^{1/\alpha}A \equiv \bar{P}_d(A).$$

[7] The trivial steady state, $\bar{L} = 0$, is unstable. Thus, given that $L_0 > 0$, this equilibrium will not be an absorbing state for the population dynamics.

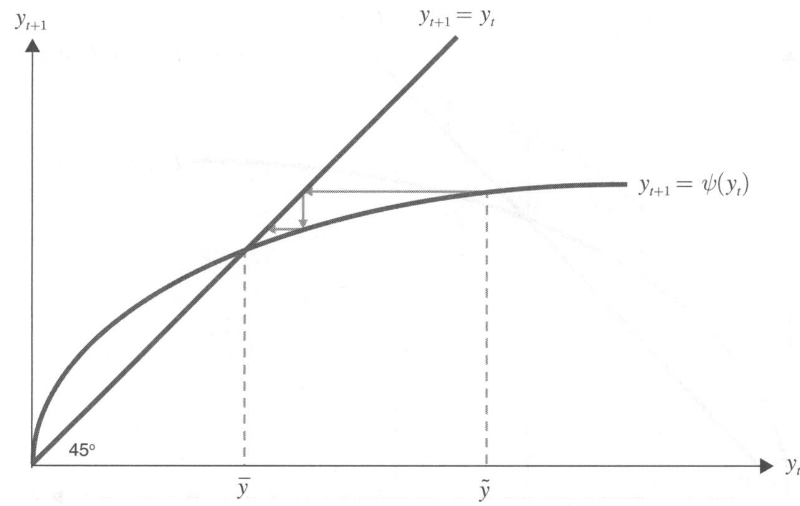

FIGURE 2. THE EVOLUTION OF INCOME PER WORKER

Importantly, as is evident from (8) and (9), an improvement in the technological environment, A, increases the steady-state levels of the adult population, $\bar{L}$, and population density, $\bar{P}_d$:

$$(10) \qquad \frac{\partial \bar{L}}{\partial A} > 0 \quad \text{and} \quad \frac{\partial \bar{P}_d}{\partial A} > 0.$$

As depicted in Figure 1, if the economy is in a steady-state equilibrium, an increase in the technological level from A to A^h generates a transition process in which population gradually increases from its initial steady-state level, $\bar{L}$, to a higher one, $\bar{L}^h$. Similarly, a decline in the population due to an epidemic such as the Black Death (1348–1350 CE) would temporarily reduce population, while temporarily increasing income per capita. The rise in income per capita, however, will generate a gradual increase in population back to the initial steady-state level, $\bar{L}$.

The Time Path of Income per Worker.—The evolution of income per worker is determined by the initial level of income per worker and the number of (surviving) children per adult. Specifically, income per worker in period $t + 1$, y_{t+1}, noting (2) and (6), is

$$(11) \qquad y_{t+1} = [(AX)/L_{t+1}]^{\alpha} = [(AX)/n_t L_t]^{\alpha} = y_t/n_t^{\alpha}.$$

Substituting (5) into (11), the time path of income per worker is governed by the first-order difference equation

$$(12) \qquad y_{t+1} = (\rho/\gamma)^{\alpha} y_t^{1-\alpha} \equiv \psi(y_t),$$

where, as depicted in Figure 2, $\psi'(y_t) > 0$ and $\psi''(y_t) < 0$, so $\psi(y_t)$ is strictly concave, and $\psi(0) = 0$, $\lim_{y_t \to 0} \psi'(y_t) = \infty$ and $\lim_{y_t \to \infty} \psi'(y_t) = 0$.

Hence, given $y_0 > 0$, there exists a unique, stable steady-state level of income per worker, $\bar{y}$:[8]

$$\bar{y} = (\rho/\gamma). \tag{13}$$

Importantly, as is evident from (2) and (13), while an advancement in the level of technology, A, increases the level of income per worker in the short run, y_t, it does not affect the steady-state level of income per worker, $\bar{y}$:

$$\frac{\partial y_t}{\partial A} > 0 \quad \text{and} \quad \frac{\partial \bar{y}}{\partial A} = 0. \tag{14}$$

As depicted in Figures 1 and 2, if the economy is in a steady-state equilibrium, an increase in the technological level from A^l to A^h generates a transition process in which income per worker initially increases to a higher level, $\tilde{y}$, reflecting higher labor productivity in the absence of population adjustment. However, as population increases, income per worker gradually declines to the initial steady-state equilibrium, $\bar{y}$. Similarly, a decline in the population due to an epidemic such as the Black Death would temporarily reduce population to $\tilde{L}$, while temporarily increasing income per capita to $\tilde{y}$. The rise in income per worker will generate a gradual increase in population back to the steady-state level, $\bar{L}$, and thus a gradual decline in income per worker back to $\bar{y}$.

C. *Testable Predictions*

The Malthusian theory generates the following testable predictions:

(i) Within a country, an increase in productivity would lead in the long run to a larger population, without altering the long-run level of income per capita.

(ii) Across countries, those characterized by superior land productivity or a superior level of technology would have, all else equal, higher population densities in the long run, but their standards of living would not reflect the degree of their technological advancement.

These predictions emerge from a Malthusian model as long as the model is based upon two fundamental features: (i) a positive effect of the standard of living on population growth, and (ii) decreasing returns to labor due to the presence of a fixed factor of production—land.[9]

[8] The trivial steady state, $\bar{y} = 0$, is unstable. Thus, given that $y_0 > 0$, this equilibrium will not be an absorbing state for the income dynamics.

[9] Specifically, these predictions would arise in the presence of a dynastic representative agent Malthusian framework (Lucas 2002), a reduced-form Malthusian-Boserupian interaction between population size and productivity growth (Kremer 1993), exogenous technological progress (Hansen and Prescott 2002), and endogenous technological progress that reflects the positive impact of population size on aggregate productivity (Galor and Weil 2000).

II. Empirical Framework

A. *Empirical Strategy*

The empirical examination of the central hypothesis of the Malthusian theory exploits exogenous sources of cross-country variation in land productivity and technological levels to examine their hypothesized *differential* effects on population density and income per capita during the time period 1–1500 CE.

In light of the potential endogeneity of population and technological progress, this research develops a novel identification strategy to examine the hypothesized effects of technological advancement on population density and income per capita. First, it establishes that the onset of the Neolithic Revolution, which marked the transition of societies from hunting and gathering to agriculture as early as 10,000 years ago, triggered a sequence of technological advancements that had a significant effect on the level of technology in the Middle Ages. As argued by Diamond (1997), an earlier onset of the Neolithic Revolution has been associated with a developmental head start that enabled the rise of a non-food-producing class whose members were essential for the advancement of written language, science, and technology, and for the formation of cities, technology-based military powers, and nation states.[10] Thus, variation in the onset of the Neolithic Revolution across the globe is exploited as a proxy for variation in the level of technological advancement during the time period 1–1500 CE.

In addition, to address the possibility that the relationship between the timing of the Neolithic transition and population density in the Common Era may itself be spurious, being perhaps codetermined by an unobserved channel such as human capital, the analysis appeals to the role of prehistoric biogeographical endowments in determining the timing of the Neolithic Revolution. Importantly, the productivity of land for agriculture in the Common Era is largely independent of the initial geographical and biogeographical endowments that were conducive for the onset of the Neolithic Revolution. While agriculture originated in regions of the world to which the most valuable domesticable wild plant and animal species were native, other regions proved more fertile and climatically favorable once the diffusion of agricultural practices brought the domesticated varieties to them (Diamond 1997). Thus, the analysis adopts an instrumental variables (IV) strategy, exploiting variation in the numbers of prehistoric domesticable species of plants and animals that were native to a region *prior to* the onset of sedentary agricultural practices as exogenous sources of variation for the number of years elapsed since the Neolithic Revolution to demonstrate its causal effect on population density in the Common Era.[11]

[10] See also Weisdorf (2005, 2009). In the context of the Malthusian model presented earlier, the Neolithic Revolution should be viewed as a large positive shock to the level of technology, A, followed by a long series of incremental aftershocks. Thus, at any given point in time, a society that experienced the Neolithic Revolution earlier would have a longer history of these aftershocks and would therefore reflect a larger steady-state population size (or, equivalently, a higher steady-state population density).

[11] The insufficient number of observations arising from the greater paucity of historical income data, as compared to data on population density, does not permit a similar instrumental variables strategy to be pursued when examining the impact of the timing of the Neolithic Revolution on income per capita. In particular, since most of the cross-sectional variation in the numbers of prehistoric domesticable species of wild plants and animals, as reported by Ola Olsson and Douglas A. Hibbs, Jr. (2005), occurs between regions rather than within regions, the

Moreover, a direct, period-specific measure of technological sophistication is also employed as an alternative metric of the level of aggregate productivity to demonstrate the qualitative robustness of the baseline results for the years 1000 CE and 1 CE.[12] Once again, the link running from the exogenous prehistoric biogeographical endowments to the level of technological advancement in the Common Era, via the timing of the Neolithic transition, enables the analysis to exploit the aforementioned biogeographical variables as instruments for the indices of technological sophistication in the years 1000 CE and 1 CE to establish their causal effects on population density in these periods.

Finally, in order to ensure that the results from the level regressions are not driven by unobserved time-invariant country fixed effects, this research also employs a first-difference estimation strategy with a lagged explanatory variable. In particular, the robustness analysis exploits cross-country variation in the change in the level of technological sophistication between the years 1000 BCE and 1 CE to explain the cross-country variations in the change in population density and the change in income per capita over the 1–1000 CE time horizon.

B. *The Data*

The most comprehensive worldwide cross-country historical estimates of population and income per capita since the year 1 CE have been assembled by Colin McEvedy and Richard Jones (1978) and Angus Maddison (2003), respectively.[13] Indeed, despite inherent problems of measurement associated with historical data, these sources remain unparalleled in providing comparable estimates across countries in the last 2,000 years and have, therefore, widely been regarded as standard sources for such data in the long-run growth literature.[14] For the purposes of the current analysis, the population density of a country for a given year is computed as population in that year, as reported by McEvedy and Jones (1978), divided by total land area.

The measure of land productivity employed is the first principal component of the percentage of arable land and an index reflecting the overall suitability of land

small sample size imposed by the availability of historical income data results in an insufficient amount of variation in explanatory variables for the first-stage regressions.

[12] The absence of sufficient variation in the underlying data obtained from Peter N. Peregrine (2003) prevents the construction of a corresponding technology measure for the year 1500 CE.

[13] It is important to note that, while the urbanization rate in 1500 CE has sometimes been used as an indicator of preindustrial economic development, it is not an alternative measure for income per capita. As suggested by the Malthusian hypothesis, technologically advanced economies have higher population densities and may thus be more urbanized, but the extent of urbanization has little or no bearing on the standard of living in the long run—it is largely a reflection of the level of technological sophistication. Indeed, the results in this study are qualitatively unaffected, particularly with respect to the impact of technological levels (as proxied by the timing of the Neolithic Revolution), when the urbanization rate in 1500 CE is used in lieu of population density as the outcome variable.

[14] Nevertheless, in the context of the current study, the use of Maddison's (2003) income per capita data could have posed a significant hurdle if the data had in part been imputed with a Malthusian viewpoint of the preindustrial world in mind. While Maddison (2008) suggests that this is not the case, the empirical investigation to follow performs a rigorous analysis to demonstrate that the baseline results remain robust under alternative specifications designed to address this particular concern surrounding Maddison's income per capita estimates. Regarding the historical population data from McEvedy and Jones (1978), while some of their estimates remain controversial, particularly those for sub-Saharan Africa and pre-Columbian Mesoamerica, a recent assessment (see, e.g., *www.census.gov/ipc/www/worldhis.html*) conducted by the US Census Bureau finds that their aggregate estimates indeed compare favorably with those obtained from other studies. Moreover, the regional estimates of McEvedy and Jones are also very similar to those presented in the more recent study by Massimo Livi-Bacci (2001).

for agriculture, based on geospatial soil quality and temperature data, as reported by Navin Ramankutty et al. (2002) and aggregated to the country level by Stelios Michalopoulos (2008).[15] The variable for the timing of the Neolithic Revolution, constructed by Louis Putterman (2008), measures the number of thousand years elapsed, relative to the year 2000 CE, since the majority of the population residing within a country's modern national borders began practicing sedentary agriculture as the primary mode of subsistence.

The index of technological sophistication is constructed based on historical cross-cultural technology data, reported with global coverage in Peregrine's (2003) *Atlas of Cultural Evolution*. In particular, for a given time period and for a given culture in the archaeological record, the *Atlas of Cultural Evolution* draws on various anthropological and historical sources to report the level of technological advancement, on a three-point scale, in each of four sectors of the economy, including communications, industry (i.e., ceramics and metallurgy), transportation, and agriculture. The index of technological sophistication is constructed following the aggregation methodology of Diego A. Comin, William Easterly, and Erick Gong (2008).[16]

C. *The Neolithic Revolution and Technological Advancement*

This section establishes that the Neolithic Revolution triggered a cumulative process of economic development, conferring a developmental head start to societies that experienced the agricultural transition earlier. In line with this assertion, Table 1 reveals preliminary results indicating that an earlier onset of the Neolithic Revolution is indeed positively and significantly correlated with the level of technological sophistication in nonagricultural sectors of the economy in the years 1000 CE and 1 CE. For instance, the coefficient estimates for the year 1000 CE, all of which are statistically significant at the 1 percent level, indicate that a 1 percent increase in the number of years elapsed since the onset of the Neolithic Revolution is associated with an increase in the level of technological advancement in the communications, industrial, and transportation sectors by 0.37, 0.07, and 0.38 percent, respectively.

These findings lend credence to the empirical strategy employed by this research to test the Malthusian theory. Specifically, they provide evidence justifying the use of the exogenous source of cross-country variation in the timing of the Neolithic Revolution as a proxy for the variation in the level of technological advancement across countries during the agricultural stage of development. Moreover, they serve as an internal consistency check between the cross-country Neolithic transition-timing variable and those on historical levels of technological sophistication, all of

[15] The use of contemporary measures of land productivity necessitates an identifying assumption that the spatial distribution of factors governing the productivity of land for agriculture has not changed significantly in the past 2,000 years. In this regard, it is important to note that the analysis at hand exploits worldwide variation in such factors, which changes dramatically only in geological time. Hence, while the assumption may not necessarily hold at a subregional level in some cases (e.g., in regions south of the Sahara where the desert has been known to be expanding gradually in the past few centuries), it is unlikely that the moments of the *global* spatial distribution of land productivity are significantly different today than they were two millennia ago. Moreover, the stability of the results over the 1–1500 CE time horizon further alleviates this potential concern.

[16] For descriptive statistics, as well as the definitions and sources of all the primary and control variables employed by the analysis, see the online Appendix.

TABLE 1—THE NEOLITHIC REVOLUTION AS A PROXY FOR TECHNOLOGICAL ADVANCEMENT

Dependent variable is level of:	Log communications technology in:		Log industrial technology in:		Log transportation technology in:	
	1000 CE	1 CE	1000 CE	1 CE	1000 CE	1 CE
	OLS (1)	OLS (2)	OLS (3)	OLS (4)	OLS (5)	OLS (6)
Log years since Neolithic transition	0.368***	0.283***	0.074***	0.068***	0.380***	0.367***
	(0.028)	(0.030)	(0.014)	(0.015)	(0.029)	(0.031)
Observations	143	143	143	143	143	143
R^2	0.48	0.26	0.17	0.12	0.52	0.51

Notes: This table demonstrates that the timing of the Neolithic Revolution is positively and significantly correlated with the level of technology in multiple nonagricultural sectors of an economy in the years 1000 CE and 1 CE. The level of technology in communications is indexed according to the absence of both true writing and mnemonic or nonwritten records, the presence of only mnemonic or nonwritten records, or the presence of both. The level of technology in industry is indexed according to the absence of both metalworks and pottery, the presence of only pottery, or the presence of both. The level of technology in transportation is indexed according to the absence of both vehicles and pack or draft animals, the presence of only pack or draft animals, or the presence of both. Robust standard error estimates are reported in parentheses.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

which are relatively new in terms of their application in the empirical literature on long-run development.

D. *The Basic Regression Model*

Formally, the baseline specifications adopted to test the Malthusian predictions regarding the effects of land productivity and the level of technological advancement on population density and income per capita are

$$\ln P_{i,t} = \alpha_{0,t} + \alpha_{1,t} \ln T_i + \alpha_{2,t} \ln X_i + \boldsymbol{\alpha}'_{3,t} \boldsymbol{\Gamma}_i + \boldsymbol{\alpha}'_{4,t} \mathbf{D}_i + \delta_{i,t}, \tag{15}$$

$$\ln y_{i,t} = \beta_{0,t} + \beta_{1,t} \ln T_i + \beta_{2,t} \ln X_i + \boldsymbol{\beta}'_{3,t} \boldsymbol{\Gamma}_i + \boldsymbol{\beta}'_{4,t} \mathbf{D}_i + \varepsilon_{i,t}, \tag{16}$$

where $P_{i,t}$ is the population density of country i in year t; $y_{i,t}$ is country i's income per capita in year t; T_i is the number of years elapsed since the onset of agriculture in country i; X_i is a measure of land productivity for country i, based on the percentage of arable land and an index of agricultural suitability; $\boldsymbol{\Gamma}_i$ is a vector of geographical controls for country i, including absolute latitude and variables gauging access to waterways; $\mathbf{D}_i$ is a vector of continental dummies; and, $\delta_{i,t}$ and $\varepsilon_{i,t}$ are country-specific disturbance terms for population density and income per capita, respectively, in year t.

III. Cross-Country Evidence

Consistent with the predictions of the Malthusian theory, the results demonstrate highly statistically significant positive effects of land productivity and the number of years elapsed since the Neolithic Revolution on population density in the years 1500 CE, 1000 CE, and 1 CE. The effects of these explanatory channels on income per capita in the corresponding periods, however, are not significantly different from zero, a result that fully complies with Malthusian priors. These results are shown to be robust to controls for other geographical factors, including absolute latitude, access to waterways, distance to the nearest technological frontier, the percentage of land in tropical versus temperate climatic zones, and small island and landlocked dummies, all of which may have had an impact on aggregate productivity either directly, by affecting the productivity of land, or indirectly by affecting trade and the diffusion of technologies.[17] Moreover, as foreshadowed by the initial findings in Table 1, the results are qualitatively unaffected when the index of technological sophistication, rather than the number of years elapsed since the Neolithic Revolution, is employed as a proxy for the level of aggregate productivity.

A. *Population Density in 1500 CE*

This section establishes the significant positive effects of land productivity and the level of technological advancement, as proxied by the timing of the Neolithic Revolution, on population density in the year 1500 CE. The results from regressions explaining log population density in the year 1500 CE are presented in Table 2. In particular, a number of specifications comprising different subsets of the explanatory variables in equation (15) are estimated to examine the independent and combined effects of the transition-timing and land-productivity channels, while controlling for other geographical factors and continental fixed effects.

Consistent with Malthusian predictions, column 1 reveals the positive relationship between log years since transition and log population density in the year 1500 CE, while controlling for continental fixed effects.[18] Specifically, the estimated OLS coefficient implies that a 1 percent increase in the number of years elapsed since the Neolithic transition increases population density in 1500 CE by 0.83 percent, an effect that is statistically significant at the 1 percent level.[19] Moreover, based on the R^2 of the regression, the transition-timing channel appears to explain 40 percent of

[17] The online Appendix presents additional findings demonstrating robustness. Specifically, it establishes that the results for population density and income per capita in 1500 CE are robust under two alternative specifications that relax potential constraints imposed by the baseline regression models, including (i) the treatment of the Americas as a single entity in accounting for continental fixed effects, and (ii) the employment of only the *common variation* in (the logs of) the percentage of arable land and the index of agricultural suitability when accounting for the effect of the land-productivity channel by way of the first principal component of these two variables. Moreover, given that historical population estimates are also available from Maddison (2003), albeit for a smaller set of countries than McEvedy and Jones (1978), the online Appendix demonstrates that the baseline results for population density in the three historical periods, obtained using data from McEvedy and Jones, are indeed qualitatively unchanged under Maddison's alternative population estimates. Finally, given the possibility that the disturbance terms in the baseline regression models may be nonspherical in nature, particularly since economic development has been spatially clustered in certain regions of the world, the online Appendix presents results from repeating the baseline analyses for population density and income per capita in the three historical periods, with the standard errors of the point estimates corrected for spatial autocorrelation following the methodology of Timothy G. Conley (1999).

[18] The results presented throughout are robust to the omission of continental dummies from the regression specifications. Without continental fixed effects, the coefficient of interest in column 1 is 1.294 [0.169], with the standard error (in brackets) indicating statistical significance at the 1 percent level.

[19] Evaluating these percentage changes at the sample means of 4,877.89 for years since transition and 6.06 for population density in 1500 CE implies that an earlier onset of the Neolithic Revolution by about 500 years is associated with an increase in population density in 1500 CE by 0.5 persons per square kilometer.

Table 2—Explaining Population Density in 1500 CE

Dependent variable is log population density in 1500 CE	OLS (1)	OLS (2)	OLS (3)	OLS (4)	OLS (5)	IV (6)
Log years since Neolithic transition	0.833***		1.025***	1.087***	1.389***	2.077***
	(0.298)		(0.223)	(0.184)	(0.224)	(0.391)
Log land productivity		0.587***	0.641***	0.576***	0.573***	0.571***
		(0.071)	(0.059)	(0.052)	(0.095)	(0.082)
Log absolute latitude		−0.425***	−0.353***	−0.314***	−0.278**	−0.248**
		(0.124)	(0.104)	(0.103)	(0.131)	(0.117)
Mean distance to nearest coast or river				−0.392***	0.220	0.250
				(0.142)	(0.346)	(0.333)
Percentage of land within 100 km of coast or river				0.899***	1.185***	1.350***
				(0.282)	(0.377)	(0.380)
Continent dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	147	147	147	147	96	96
R^2	0.40	0.60	0.66	0.73	0.73	0.70
First-stage F-statistic	–	–	–	–	–	14.65
Overidentifying restrictions p-value	–	–	–	–	–	0.440

Notes: This table establishes, consistently with Malthusian predictions, the significant positive effects of land productivity, and the level of technological advancement, as proxied by the timing of the Neolithic Revolution, on population density in the year 1500 CE, while controlling for access to navigable waterways, absolute latitude, and unobserved continental fixed effects. Log land productivity is the first principal component of the log of the percentage of arable land and the log of an agricultural suitability index. The IV regression employs the numbers of prehistoric domesticable species of plants and animals as instruments for log transition timing. The statistic for the first-stage F-test of these instruments is significant at the 1 percent level. The p-value for the overidentifying restrictions test corresponds to Hansen's J statistic, distributed in this case as χ^2 with one degree of freedom. A single continent dummy is used to represent the Americas, which is natural given the historical period examined. Regressions (5)–(6) do not employ the Oceania dummy due to a single observation for this continent in the IV data-restricted sample. Robust standard error estimates are reported in parentheses.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

the variation in log population density in 1500 CE along with the dummies capturing unobserved continental characteristics.

The effect of the land-productivity channel, controlling for absolute latitude and continental fixed effects, is reported in column 2. In line with theoretical predictions, a 1 percent increase in land productivity raises population density in 1500 CE by 0.59 percent, an effect that is also significant at the 1 percent level. Interestingly, in contrast to the relationship between absolute latitude and contemporary income per capita, the estimated elasticity of population density in 1500 CE with respect to absolute latitude suggests that economic development during this period was on average higher at latitudinal bands closer to the equator.[20] Thus, while proximity to the equator was beneficial in the agricultural stage of development, it appears detrimental in the

[20] An interesting potential explanation for this finding comes from an admittedly contested hypothesis in the field of evolutionary ecology. In particular, biodiversity tends to decline as one moves farther away from the equator—a phenomenon known as *Rapoport's Rule*—due to the stronger forces of natural selection arising from wider seasonal variation in climate at higher absolute latitudes. Lower resource diversity at higher absolute latitudes would imply lower carrying capacities of these environments due to the greater extinction susceptibility of the resource base under adverse natural shocks such as disease and sudden climatic fluctuations. The lower carrying capacities of these environments would, in turn, imply lower levels of human population density.

industrial stage. The R^2 of the regression indicates that, along with continental fixed effects and absolute latitude, the land-productivity channel explains 60 percent of the cross-country variation in log population density in 1500 CE.

Column 3 presents the results from examining the combined explanatory power of the previous two regressions. The estimated coefficients on the transition-timing and land-productivity variables remain highly statistically significant and continue to retain their expected signs, while increasing slightly in magnitude in comparison to their estimates in earlier columns. Furthermore, transition timing and land productivity together explain 66 percent of the variation in log population density in 1500 CE, along with absolute latitude and continental fixed effects.

The explanatory power of the regression in column 3 improves by an additional 7 percentage points once controls for access to waterways are accounted for in column 4, which constitutes the baseline regression specification for population density in 1500 CE. In comparison to the estimates reported in column 3, the effects of the transition-timing and land-productivity variables remain reassuringly stable in both magnitude and statistical significance when subjected to the additional geographical controls. Moreover, the estimated coefficients on the additional geographical controls indicate significant effects consistent with the assertion that better access to waterways has been historically beneficial for economic development by fostering urbanization, international trade, and technology diffusion. To interpret the baseline effects of the variables of interest, a 1 percent increase in the number of years elapsed since the Neolithic Revolution raises population density in 1500 CE by 1.09 percent, conditional on land productivity, absolute latitude, waterway access, and continental fixed effects. Similarly, a 1 percent increase in land productivity generates, ceteris paribus, a 0.58 percent increase in population density in 1500 CE.[21] These conditional effects of the transition-timing and land-productivity channels from the baseline specification are depicted as partial regression lines on the scatter plots in panels A and B of Figure 3, respectively.

The analysis now turns to address issues regarding causality, particularly with respect to the transition-timing variable. Specifically, while variations in land productivity and other geographical characteristics are inarguably exogenous to the cross-country variation in population density, the onset of the Neolithic Revolution and the outcome variable of interest may in fact be endogenously determined. In particular, although reverse causality is not a source of concern, given that the vast majority of countries underwent the Neolithic transition prior to the Common Era, the OLS estimates of the effect of the time elapsed since the transition to agriculture may suffer from omitted variable bias, reflecting spurious correlations with the outcome variable being examined.

To establish the causal effect of the timing of the Neolithic transition on population density in the Common Era, the investigation appeals to Diamond's (1997) hypothesis on the role of exogenous geographical and biogeographical endowments in determining the timing of the Neolithic Revolution. Accordingly, the emergence and subsequent diffusion of agricultural practices were primarily driven by geographical

[21] In the absence of continental fixed effects, the coefficient associated with the transition-timing channel is 1.373 [0.118], while that associated with the land-productivity channel is 0.586 [0.058], with the standard errors (in brackets) indicating statistical significance at the 1 percent level.

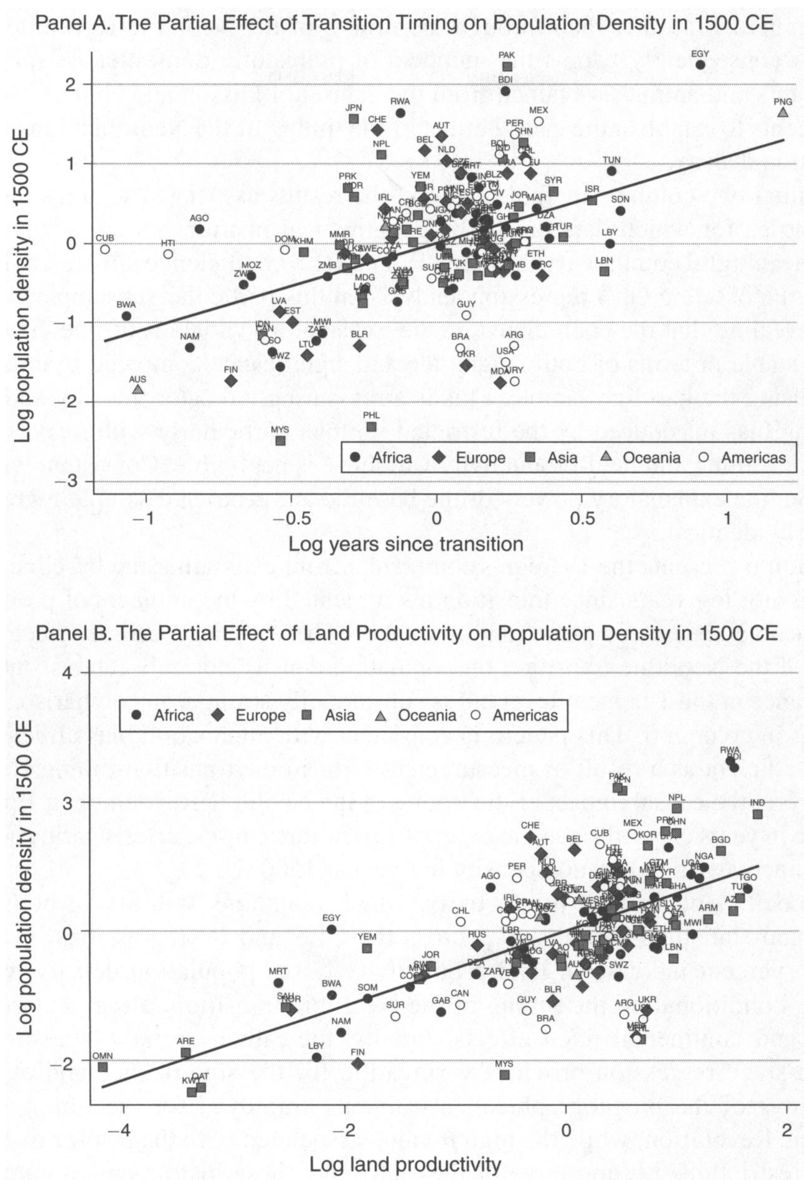

FIGURE 3. TRANSITION TIMING, LAND PRODUCTIVITY, AND POPULATION DENSITY IN 1500 CE

Notes: This figure depicts the partial regression line for the effect of transition timing (land productivity) on population density in the year 1500 CE, while controlling for the influence of land productivity (transition timing), absolute latitude, access to waterways, and continental fixed effects. Thus, the x- and y-axes plot the residuals obtained from regressing transition timing (land productivity) and population density, respectively, on the aforementioned set of covariates.

conditions such as climate, continental size and orientation, as well as the availability of wild plant and animal species amenable to domestication. However, while geographical factors certainly continued to play a direct role in economic development after the onset of agriculture, it is postulated that the availability of prehistoric domesticable wild plant and animal species did not influence population density in the Common Era other than through the timing of the Neolithic Revolution. The analysis consequently adopts the numbers of prehistoric domesticable species of wild plants and animals, obtained from the dataset of Olsson and Hibbs (2005), as instruments to establish the causal effect of the timing of the Neolithic transition on population density.

The final two columns in Table 2 report the results associated with a subsample of countries for which data on the biogeographical instruments are available. To allow meaningful comparisons between IV and OLS coefficient estimates, column 5 repeats the baseline OLS regression analysis on this particular subsample of countries, revealing that the coefficients on the explanatory variables of interest remain largely stable in terms of both magnitude and significance compared to those estimated using the baseline sample. This is a reassuring indicator that any additional sampling bias introduced by the restricted sample, particularly with respect to the transition-timing and land-productivity variables, is negligible. Consistent with this assertion, the explanatory powers of the baseline and restricted sample regressions are nearly identical.

Column 6 presents the IV regression results from estimating the baseline specification with log years since transition instrumented by the numbers of prehistoric domesticable species of plants and animals.[22] The estimated causal effect of the timing of the Neolithic transition on population density not only retains statistical significance at the 1 percent level but is substantially stronger in comparison to the estimate in column 5. This pattern is consistent with attenuation bias afflicting the OLS coefficient as a result of measurement error in the transition-timing variable. To interpret the causal impact of the timing of the Neolithic Revolution, a 1 percent increase in years elapsed since the onset of agriculture causes, ceteris paribus, a 2.08 percent increase in population density in the year 1500 CE.

The coefficient on land productivity, which maintains stability in both magnitude and statistical significance across the OLS and IV regressions, indicates that a 1 percent increase in land productivity raises population density by 0.57 percent, conditional on the timing of the Neolithic transition, other geographical factors and continental fixed effects. Finally, the rather strong F-statistic from the first-stage regression provides verification for the significance and explanatory power of the biogeographical instruments employed for the timing of the Neolithic Revolution, while the high p-value associated with the test for overidentifying restrictions is supportive of the claim that these instruments do not exert any independent influence on population density in 1500 CE other than through the transition-timing channel.

B. *Population Density in Earlier Historical Periods*

This section demonstrates the significant positive effects of land productivity and the level of technological advancement, as proxied by the timing of the Neolithic Revolution, on population density in the years 1000 CE and 1 CE. The results from regressions explaining log population density in the years 1000 CE and 1 CE are

[22] Table A.1 in the Appendix summarizes the first-stage regression results from all IV regressions examined by the current analysis.

TABLE 3—EXPLAINING POPULATION DENSITY IN 1000 CE

Dependent variable is log population density in 1000 CE	OLS (1)	OLS (2)	OLS (3)	OLS (4)	OLS (5)	IV (6)
Log years since Neolithic transition	1.232***		1.435***	1.480***	1.803***	2.933***
	(0.293)		(0.243)	(0.205)	(0.251)	(0.504)
Log land productivity		0.470***	0.555***	0.497***	0.535***	0.549***
		(0.081)	(0.065)	(0.056)	(0.098)	(0.092)
Log absolute latitude		−0.377**	−0.283**	−0.229**	−0.147	−0.095
		(0.148)	(0.116)	(0.111)	(0.127)	(0.116)
Mean distance to nearest coast or river				−0.528***	0.147	0.225
				(0.153)	(0.338)	(0.354)
Percentage of land within 100 km of coast or river				0.716**	1.050**	1.358***
				(0.323)	(0.421)	(0.465)
Continent dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	142	142	142	142	94	94
R^2	0.38	0.46	0.59	0.67	0.69	0.62
First-stage F-statistic	–	–	–	–	–	15.10
Overidentifying restrictions p-value	–	–	–	–	–	0.281

Notes: This table establishes, consistently with Malthusian predictions, the significant positive effects of land productivity and the level of technological advancement, as proxied by the timing of the Neolithic Revolution, on population density in the year 1000 CE, while controlling for access to navigable waterways, absolute latitude, and unobserved continental fixed effects. Log land productivity is the first principal component of the log of the percentage of arable land and the log of an agricultural suitability index. The IV regression employs the numbers of prehistoric domesticable species of plants and animals as instruments for log transition timing. The statistic for the first-stage F-test of these instruments is significant at the 1 percent level. The p-value for the overidentifying restrictions test corresponds to Hansen's J statistic, distributed in this case as χ^2 with one degree of freedom. A single continent dummy is used to represent the Americas, which is natural given the historical period examined. Regressions (5)–(6) do not employ the Oceania dummy due to a single observation for this continent in the IV data-restricted sample. Robust standard error estimates are reported in parentheses.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

presented in Tables 3 and 4, respectively. As before, the independent and combined explanatory powers of the transition-timing and land-productivity channels are examined while controlling for other geographical factors and unobserved continental characteristics.

In line with the empirical predictions of the Malthusian theory, the findings reveal highly statistically significant positive effects of land productivity and an earlier transition to agriculture on population density in these earlier historical periods as well. Moreover, the positive impact on economic development of geographical factors capturing better access to waterways is also confirmed for these earlier periods.[23]

[23] The inverse correlation between absolute latitude and population density is maintained in the 1000 CE analysis, but appears ambiguous in the 1 CE analysis. This pattern may, in part, reflect increasing returns associated with societies residing closer to the equator during the Malthusian stage of development. In particular, as a result of agglomeration and latitudinally specific technology diffusion, the initial advantage enjoyed by equatorial societies during the Malthusian epoch became more pronounced over time. Thus, the observed negative cross-sectional relationship between absolute latitude and population density, which is somewhat weak in the year 1 CE, becomes progressively stronger in the years 1000 CE and 1500 CE.

TABLE 4—EXPLAINING POPULATION DENSITY IN 1 CE

Dependent variable is log population density in 1 CE	OLS (1)	OLS (2)	OLS (3)	OLS (4)	OLS (5)	IV (6)
Log years since Neolithic transition	1.560***		1.903***	1.930***	2.561***	3.459***
	(0.326)		(0.312)	(0.272)	(0.369)	(0.437)
Log land productivity		0.404***	0.556***	0.394***	0.421***	0.479***
		(0.106)	(0.081)	(0.067)	(0.094)	(0.089)
Log absolute latitude		−0.080	−0.030	0.057	0.116	0.113
		(0.161)	(0.120)	(0.101)	(0.121)	(0.113)
Mean distance to nearest coast or river				−0.685***	−0.418	−0.320
				(0.155)	(0.273)	(0.306)
Percentage of land within 100 km of coast or river				0.857**	1.108***	1.360***
				(0.351)	(0.412)	(0.488)
Continent dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	128	128	128	128	83	83
R^2	0.47	0.41	0.59	0.69	0.75	0.72
First-stage F-statistic	–	–	–	–	–	10.85
Overidentifying restrictions p-value	–	–	–	–	–	0.590

Notes: This table establishes, consistently with Malthusian predictions, the significant positive effects of land productivity and the level of technological advancement, as proxied by the timing of the Neolithic Revolution, on population density in the year 1 CE, while controlling for access to navigable waterways, absolute latitude, and unobserved continental fixed effects. Log land productivity is the first principal component of the log of the percentage of arable land and the log of an agricultural suitability index. The IV regression employs the numbers of prehistoric domesticable species of plants and animals as instruments for log transition timing. The statistic for the first-stage F-test of these instruments is significant at the 1 percent level. The p-value for the overidentifying restrictions test corresponds to Hansen's J statistic, distributed in this case as χ^2 with one degree of freedom. A single continent dummy is used to represent the Americas, which is natural given the historical period examined. Regressions (5)–(6) do not employ the Oceania dummy due to a single observation for this continent in the IV data-restricted sample. Robust standard error estimates are reported in parentheses.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

The stability patterns exhibited by the magnitude and significance of the coefficients on the explanatory variables of interest in Tables 3 and 4 are strikingly similar to those observed in the 1500 CE analysis. Thus, for instance, while statistical significance remains unaffected across specifications, the independent effects of Neolithic transition timing and land productivity from the first two columns in each table increase slightly in magnitude when both channels are examined concurrently in column 3, and remain stable thereafter when subjected to the additional geographical controls in the baseline regression specification of the fourth column. This is a reassuring indicator that the variance-covariance characteristics of the regression samples employed for the different periods are not fundamentally different from one another, despite differences in sample size due to the greater unavailability of population density data in the earlier historical periods. The qualitative similarity of the results across periods also suggests that the empirical findings are indeed more plausibly associated with the Malthusian theory, as opposed to being consistently generated by spurious correlations between population density and the explanatory variables of interest across the different historical periods.

To interpret the baseline effects of interest from column 4 of the analysis for each historical period, a 1 percent increase in the number of years elapsed since the onset of the Neolithic Revolution raises population density in the years 1000 CE and 1 CE by 1.48 and 1.93 percent, respectively, conditional on the productivity of land, absolute latitude, access to waterways, and continental fixed effects.[24] Similarly, a 1 percent increase in land productivity is associated with, ceteris paribus, a 0.50 percent increase in population density in 1000 CE and a 0.39 percent increase in population density in 1 CE.[25]

For the 1000 CE analysis, the additional sampling bias introduced on OLS estimates by moving to the IV-restricted subsample in column 5 is similar to that observed earlier in Table 2, whereas the bias appears somewhat larger for the analysis in 1 CE. This is partly attributable to the smaller size of the subsample in the latter analysis. The IV regressions in column 6, however, once again reflect the pattern that the causal effect of transition timing on population density in each period is stronger than its corresponding reduced-form effect, while the effect of land productivity remains rather stable across the OLS and IV specifications. In addition, the strength and credibility of the numbers of domesticable plant and animal species as instruments continue to be supported by their explanatory power in the first-stage regressions and by the results of the overidentifying restrictions tests. The similarity of these findings with those obtained in the 1500 CE analysis reinforces the validity of these instruments and, thereby, lends further credence to the causal effect of the timing of the Neolithic transition on population density.

Finally, turning attention to the differences in coefficient estimates obtained for the three periods, it is interesting to note that, while the positive effect of land productivity on population density remains rather stable, that of the number of years elapsed since the onset of agriculture declines over time. For instance, comparing the IV coefficient estimates on the transition-timing variable across Tables 2–4, the positive causal impact of the Neolithic Revolution on population density diminishes by 0.53 percentage points over the 1–1000 CE time horizon and by 0.85 percentage points over the subsequent 500-year period. This pattern is consistently reflected by all regression specifications examining the effect of the transition-timing variable, lending support to the assertion that the process of development initiated by the technological breakthrough of the Neolithic Revolution conferred social gains characterized by diminishing returns over time.[26]

[24] In both the 1000 CE and 1 CE samples, evaluating these percentage changes at the sample means for years since transition and population density implies that an earlier onset of the Neolithic Revolution by about 500 years is associated with an increase in population density by 0.5 persons per square kilometer. Despite differences in the estimated elasticities between the two periods, the similarity of the effects at the sample means arises due to counteracting differences in the sample means themselves. Specifically, while population density in 1000 CE has a sample mean of 3.59, the mean in 1 CE is only 2.54.

[25] The online Appendix depicts these conditional effects as partial regression lines on the scatter plots in Figures W.1(a) and W.1(b) for the 1000 CE analysis, and in Figures W.2(a) and W.2(b) for the 1 CE analysis.

[26] The assertion that the process of development initiated by the Neolithic Revolution was characterized by diminishing returns over time implies that, given a sufficiently large lag following the transition, societies should be expected to converge toward a Malthusian steady state conditional on the productivity of land and other geographical factors. Hence, the cross-sectional relationship between population density and the number of years elapsed since the Neolithic transition should be expected to exhibit some concavity. This prediction was tested using the following specification:

$$\ln P_{i,t} = \theta_{0,t} + \theta_{1,t}T_i + \theta_{2,t}T_i^2 + \theta_{3,t}\ln X_i + \boldsymbol{\theta}_{4,t}'\boldsymbol{\Gamma}_i + \boldsymbol{\theta}_{5,t}'\mathbf{D}_i + \omega_{i,t}.$$

Table 5—Effects on Income per Capita versus Population Density

Dependent variable is:	Log income per capita in:			Log population density in:		
	1500 CE	1000 CE	1 CE	1500 CE	1000 CE	1 CE
	OLS	OLS	OLS	OLS	OLS	OLS
	(1)	(2)	(3)	(4)	(5)	(6)
Log years since Neolithic transition	0.159	0.073	0.109	1.337**	0.832**	1.006**
	(0.136)	(0.045)	(0.072)	(0.594)	(0.363)	(0.481)
Log land productivity	0.041	−0.021	−0.001	0.584***	0.364***	0.681**
	(0.025)	(0.025)	(0.027)	(0.159)	(0.110)	(0.255)
Log absolute latitude	−0.041	0.060	−0.175	0.050	−2.140**	−2.163**
	(0.073)	(0.147)	(0.175)	(0.463)	(0.801)	(0.979)
Mean distance to nearest coast or river	0.215	−0.111	0.043	−0.429	−0.237	0.118
	(0.198)	(0.138)	(0.159)	(1.237)	(0.751)	(0.883)
Percentage of land within 100 km of coast or river	0.124	−0.150	0.042	1.855**	1.326**	0.228
	(0.145)	(0.121)	(0.127)	(0.820)	(0.615)	(0.919)
Continent dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	31	26	29	31	26	29
R^2	0.66	0.68	0.33	0.88	0.95	0.89

Notes: This table establishes, consistently with Malthusian predictions, the relatively small effects of land productivity and the level of technological advancement, as proxied by the timing of the Neolithic Revolution, on income per capita in the years 1500 CE, 1000 CE, and 1 CE, but their significantly larger effects on population density in the same time periods, while controlling for access to navigable waterways, absolute latitude, and unobserved continental fixed effects. Log land productivity is the first principal component of the log of the percentage of arable land and the log of an agricultural suitability index. A single continent dummy is used to represent the Americas, which is natural given the historical period examined. Regressions (2)–(3) and (5)–(6) do not employ the Oceania dummy due to a single observation for this continent in the corresponding regression samples, restricted by the availability of income per capita data. Robust standard error estimates are reported in parentheses.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

C. *Income per Capita versus Population Density*

This section examines the Malthusian prediction regarding the neutrality of the standard of living with respect to land productivity and the level of technological advancement, as proxied by the timing of the Neolithic Revolution. Table 5 presents the results from estimating the baseline empirical model, as specified in equation (16), for income per capita in the years 1500 CE, 1000 CE, and 1 CE. Since historical income per capita data are available for a relatively smaller set of countries, the analysis at hand also conducts corresponding tests for population density using the income per capita data-restricted samples for the three historical periods. This permits an impartial assessment of whether higher land productivity and an earlier onset of the Neolithic Revolution are manifested mostly in terms of higher population density, as opposed to higher income per capita, as the Malthusian theory would predict.

Consistent with the aforementioned prediction, the OLS regression for 1500 CE yields $\theta_{1,1500} = 0.630$ [0.133] and $\theta_{2,1500} = -0.033$ [0.011] with the standard errors (in brackets) indicating that both estimates are statistically significant at the 1 percent level. Moreover, in line with the prediction that a concave relationship should not necessarily be observed in an earlier period, the regression for 1 CE yields $\theta_{1,1} = 0.755$ [0.172] and $\theta_{2,1} = -0.020$ [0.013] with the standard errors indicating that the first-order (linear) effect is statistically significant at the 1 percent level, whereas the second-order (quadratic) effect is statistically insignificant.

Columns 1–3 reveal that income per capita in each historical period is effectively neutral to variations in the timing of the Neolithic Revolution, the agricultural productivity of land, and other productivity-enhancing geographical factors, conditional on continental fixed effects.[27] In particular, the effects of transition timing and land productivity on income per capita are not only substantially smaller than those on population density, they are also not statistically different from zero at conventional levels of significance.[28] Moreover, the other geographical factors, which, arguably, had facilitated trade and technology diffusion, do not appear to significantly affect income per capita.

In contrast, the regressions in columns 4–6 reveal, exploiting the same variation in explanatory variables as in the preceding income per capita regressions, that the elasticities of population density in each period with respect to Neolithic transition timing and land productivity are not only highly statistically significant, but also larger by about an order of magnitude than the corresponding elasticities of income per capita. Thus, for the year 1500 CE, a 1 percent increase in the number of years elapsed since the Neolithic Revolution raises population density by 1.34 percent but income per capita by only 0.16 percent, conditional on land productivity, geographical factors, and continental fixed effects. Similarly, a 1 percent increase in land productivity is associated, ceteris paribus, with a 0.58 percent increase in population density in 1500 CE but only a 0.04 percent increase in income per capita in the same time period. The conditional effects of Neolithic transition timing and land productivity on income per capita versus population density in the year 1500 CE are depicted as partial regression lines on the scatter plots in panels A and B of Figure 4 for income per capita, and in panels A and B of Figure 5 for population density.

While the results revealing the cross-country neutrality of income per capita, despite differences in aggregate productivity, are fully consistent with Malthusian predictions, there may exist potential concerns regarding the quality of the income per capita data employed by the current analysis. In particular, contrary to Maddison's (2008) implicit assertion, if the historical income per capita estimates were in part imputed under the Malthusian prior regarding similarities in the standard of living across countries, then applying these data to test the Malthusian theory itself would clearly be invalid.[29]

[27] The rather high R^2 associated with each of these regressions is due to the inclusion of continental fixed effects in the specification.

[28] Although Putterman (2008) reports a positive and significant effect of transition timing on income per capita in the year 1500 CE, this finding is, in fact, entirely spurious. Specifically, the relationship reported by Putterman disappears (i.e., the coefficient on transition timing is nearly zero and statistically insignificant) once continental fixed effects are added to the regression.

[29] A closer look at some properties of Maddison's (2003) data suggests that this need not be a concern. Figure W.3, presented in the online Appendix, depicts the cross-sectional variability of income per capita according to Maddison's estimates for the year 1500 CE, plotting the cumulative distribution of income per capita against quantiles of the data. The 45-degree line in the figure therefore corresponds to a uniform distribution, wherein each observation would possess a unique value for income per capita. Indeed, the close proximity of Maddison's observations to the 45-degree line indicates a healthy degree of variability across countries, suggesting that the data were not conditioned to conform to a Malthusian view of the world. Moreover, Figure W.4, illustrating the intertemporal variability of income per capita over the 1000–1500 CE time horizon, provides further assurance that Maddison's estimates are not tainted by implicit assumptions that make the data unreliable for testing the Malthusian theory. In particular, the departure of the vast majority of observations from the 45-degree line in the figure is at odds with an unconditional Malthusian prior that would otherwise necessitate stagnation in income per capita over time, and hence require a greater proximity of observations to the 45-degree line.

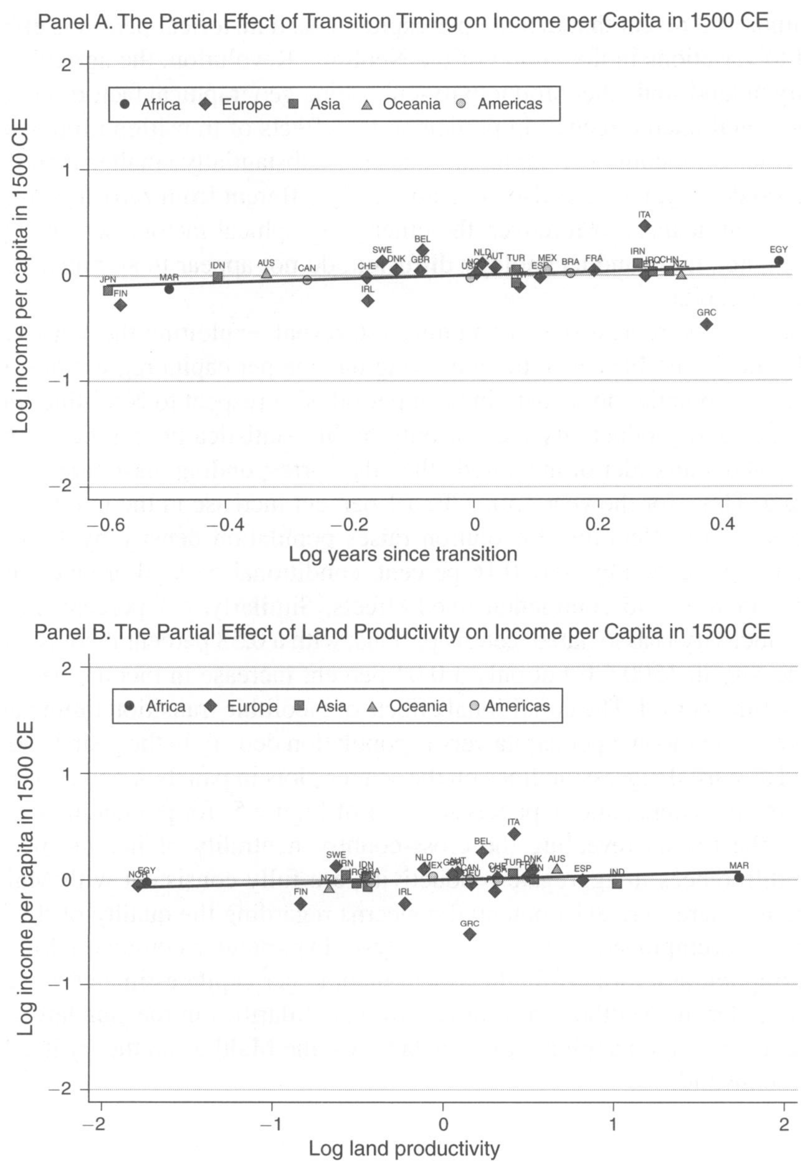

Figure 4. Transition Timing, Land Productivity, and Income per Capita in 1500 CE

Notes: This figure depicts the partial regression line for the effect of transition timing (land productivity) on income per capita in the year 1500 CE, while controlling for the influence of land productivity (transition timing), absolute latitude, access to waterways, and continental fixed effects. Thus, the x- and y-axes plot the residuals obtained from regressing transition timing (land productivity) and income per capita, respectively, on the aforementioned set of covariates.

The current investigation therefore performs a rigorous robustness analysis of the baseline results with respect to the aforementioned data quality concerns. In particular, columns 1–3 in Table 6 reveal the results from estimating the baseline specification for income per capita in the three historical periods, using regressions where each observation is weighted *down* according to the number of observations in the sample

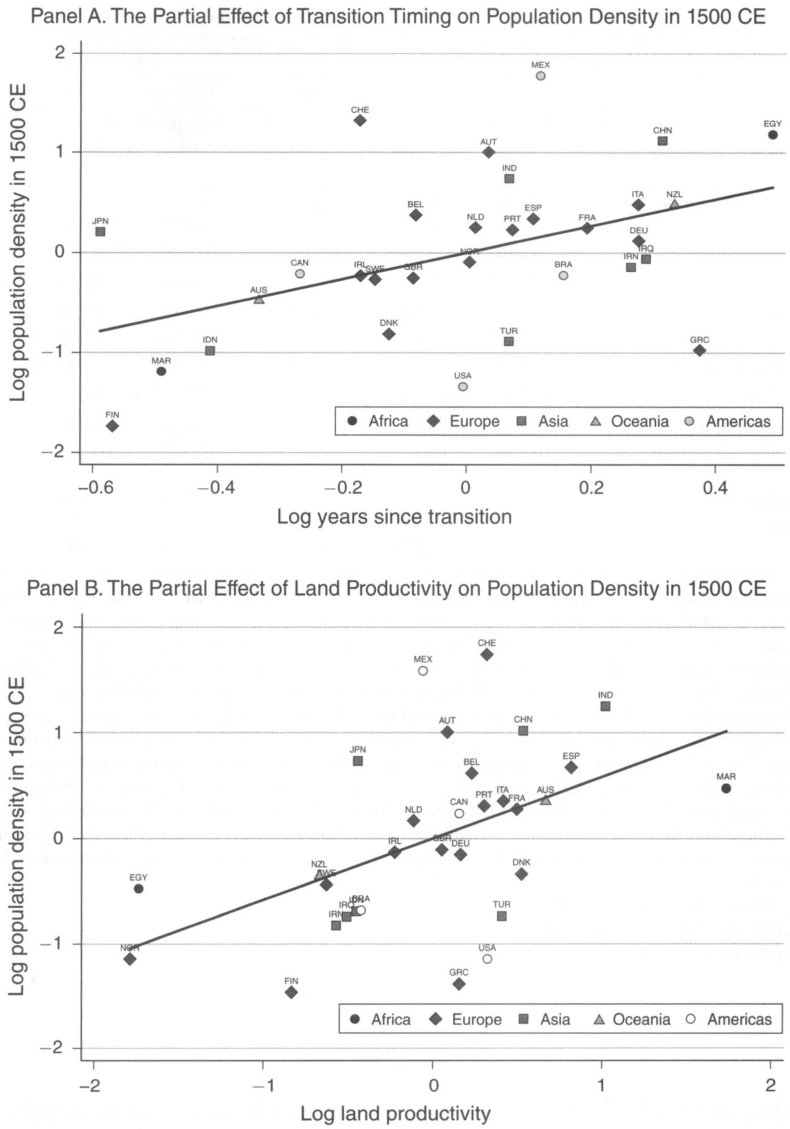

FIGURE 5. TRANSITION TIMING, LAND PRODUCTIVITY, AND POPULATION DENSITY IN 1500 CE

Notes: This figure depicts, *using the income per capita data-restricted sample*, the partial regression line for the effect of transition timing (land productivity) on population density in the year 1500 CE, while controlling for the influence of land productivity (transition timing), absolute latitude, access to waterways, and continental fixed effects. Thus, the x- and y-axes plot the residuals obtained from regressing transition timing (land productivity) and population density, respectively, on the aforementioned set of covariates.

reported to possess the same level of income per capita as the observation in question.[30] To the extent that the potential lack of variability in subsets of Maddison's income per data may have biased the baseline results in favor of the Malthusian theory,

[30]The notes to Table 6 provide more formal details on the sample weighting methodologies.

Table 6—Robustness to Income per Capita Data Quality Concerns

Observations weighted according to:	Income data frequency			Total population size		
	1500 CE	1000 CE	1 CE	1500 CE	1000 CE	1 CE
Dependent variable is log income per capita in:	Weighted OLS (1)	Weighted OLS (2)	Weighted OLS (3)	Weighted OLS (4)	Weighted OLS (5)	Weighted OLS (6)
Log years since Neolithic transition	0.173	0.122*	0.189	0.278	0.143*	0.289
	(0.162)	(0.063)	(0.121)	(0.171)	(0.068)	(0.175)
Log land productivity	0.039	−0.045*	0.008	−0.005	−0.062*	−0.011
	(0.023)	(0.022)	(0.031)	(0.026)	(0.030)	(0.027)
Log absolute latitude	−0.042	0.205*	−0.442	−0.089	0.298***	0.080
	(0.080)	(0.108)	(0.362)	(0.052)	(0.031)	(0.089)
Mean distance to nearest coast or river	0.219	−0.370**	0.139	0.332**	−0.592***	−0.180
	(0.202)	(0.148)	(0.298)	(0.148)	(0.108)	(0.189)
Percentage of land within 100 km of coast or river	0.153	−0.228	0.159	0.329	−0.477***	0.003
	(0.169)	(0.137)	(0.257)	(0.227)	(0.122)	(0.277)
Continent dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	31	26	29	31	26	29
R^2	0.54	0.79	0.29	0.74	0.83	0.45

Notes: This table demonstrates that the relatively small effects of land productivity and the level of technological advancement, as proxied by the timing of the Neolithic Revolution, on income per capita in the years 1500 CE, 1000 CE, and 1 CE remain robust under two different weighted regression methodologies, designed to dispel concerns regarding the quality of the historical income per capita data series. Log land productivity is the first principal component of the log of the percentage of arable land and the log of an agricultural suitability index. The weight of country i in regressions (1)–(3) is *inversely* proportional to the frequency with which i's income per capita occurs in the corresponding samples, i.e., $w_i = n_i^{-1}/\sum_i n_i^{-1}$, where n_i is the number of countries with income per capita identical to i. The weight of country i in regressions (4)–(6) is *directly* proportional to the population size of i in the corresponding samples, i.e., $w_i = p_i/\sum_i p_i$, where p_i is the size of the population of i. A single continent dummy is used to represent the Americas, which is natural given the historical period examined. Regressions (2)–(3) and (5)–(6) do not employ the Oceania dummy due to a single observation for this continent in the corresponding regression samples, restricted by the availability of income per capita data. Robust standard error estimates are reported in parentheses.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

this methodology alleviates such bias in the regression by reducing the relative importance of clusters of the data where observed variation is lacking.

A comparison of each of the first three columns between Tables 5 and 6 indicates that the baseline results remain both quantitatively and qualitatively robust with respect to the aforementioned weighting procedure. The quantitative robustness of the results are verified by the fact that, despite the statistical significance of some of the effects in the year 1000 CE under the weighted methodology, the transition-timing and land-productivity channels continue to remain economically non-substantial for income per capita in all three periods, as reflected by estimated elasticities that are still about an order of magnitude smaller than those of population density in the corresponding periods.

Reassuringly, a similar robustness pattern of the baseline results for income per capita is observed with respect to columns 4–6 of Table 6 where an alternative sample weighting procedure is employed, with individual observations weighted *up* according to their respective population densities. To the extent that the sample variation in income per capita may have been artificially introduced under the premise that technologically advanced societies, as reflected by their higher population densities, also enjoyed *marginally* higher standards of living, this weighting procedure would *a priori* amplify the manifestation of technological differences as differences in income per capita, and thus bias the results against Malthusian predictions. Nevertheless, despite exacerbating any systematic bias in favor of rejecting the theory, the results obtained under this weighting procedure continue to demonstrate the insignificance of the land-productivity and transition-timing channels for income per capita in all three historical periods.

To summarize the main findings of the analysis thus far, the results indicate that more productive societies sustained higher population densities, as opposed to higher standards of living, during the time period 1–1500 CE. These findings are entirely consistent with the Malthusian prediction that in preindustrial economies, resources temporarily generated by more productive technological environments were ultimately channeled into population growth, with negligible long-run effects on income per capita.

D. *Technological Sophistication*

This section demonstrates the qualitative robustness of the results, regarding the significant positive effect of technology, as proxied by the timing of the Neolithic Revolution, on population density, but its neutrality for income per capita, under direct measures of technological advancement. In particular, Table 7 presents the findings from estimating the baseline specification for population density and income per capita in the years 1000 CE and 1 CE, employing the index of technological sophistication corresponding to these periods, in lieu of the number of years elapsed since the Neolithic Revolution, as an indicator of the level of aggregate productivity.

As mentioned previously, the index of technological sophistication in each period is based on cross-cultural, sector-specific technology data from Peregrine (2003), aggregated up to the country level by averaging across sectors and cultures within a country, following the aggregation methodology of Comin, Easterly, and Gong (2008). Specifically, the index not only captures the level of technological advancement in communications, transportation, and industry, but also incorporates information on the prevalence of sedentary agricultural practices relative to hunting and gathering.[31] Since the timing of the Neolithic transition is *a priori* expected to be highly correlated with the prevalence of agriculture across countries in both 1000 CE and 1 CE, its inclusion as an explanatory variable in the current analysis would constitute the exploitation of redundant information and potentially obfuscate the results of the analysis. The regressions in Table 7 therefore omit the timing of the Neolithic Revolution as an explanatory variable for both population density and income per capita in the two periods examined.[32]

[31] See the online Appendix for additional details.

[32] Consistent with the symptoms of multicollinearity, the inclusion of the transition-timing variable in these regressions results in the coefficients of interest possessing larger standard errors with relatively minor effects on the coefficient magnitudes themselves.

TABLE 7—ROBUSTNESS TO DIRECT MEASURES OF TECHNOLOGICAL SOPHISTICATION

Dependent variable is:	Log population density in:		Log income per capita in:		Log population density in:	
	1000 CE	1 CE	1000 CE	1 CE	1000 CE	1 CE
	OLS full sample (1)	OLS full sample (2)	OLS income sample (3)	OLS income sample (4)	OLS income sample (5)	OLS income sample (6)
Log technology index in relevant period	4.315***	4.216***	0.064	0.678	12.762***	7.461**
	(0.850)	(0.745)	(0.230)	(0.432)	(0.918)	(3.181)
Log land productivity	0.449***	0.379***	−0.016	0.004	0.429**	0.725**
	(0.056)	(0.082)	(0.030)	(0.033)	(0.182)	(0.303)
Log absolute latitude	−0.283**	−0.051	0.036	−0.198	−1.919***	−2.350***
	(0.120)	(0.127)	(0.161)	(0.176)	(0.576)	(0.784)
Mean distance to nearest coast or river	−0.638***	−0.782***	−0.092	0.114	0.609	0.886
	(0.188)	(0.198)	(0.144)	(0.164)	(0.469)	(0.904)
Percentage of land within 100 km of coast or river	0.385	0.237	−0.156	0.092	1.265**	0.788
	(0.313)	(0.329)	(0.139)	(0.136)	(0.555)	(0.934)
Continent dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	140	129	26	29	26	29
R^2	0.61	0.62	0.64	0.30	0.97	0.88

Notes: This table demonstrates that the relatively small effect of the level of technological advancement on income per capita in the years 1000 CE and 1 CE, but its significantly larger effect on population density in the same time periods, remains qualitatively robust when direct measures of technological sophistication for the corresponding years are used in lieu of the timing of the Neolithic Revolution. The technology index for a given time period reflects the average degree of technological sophistication across communications, transportation, industrial, and agricultural sectors in that period. The almost perfect collinearity between the degree of technological sophistication in the agricultural sector and the timing of the Neolithic transition does not permit the use of the latter as a covariate in these regressions. Log land productivity is the first principal component of the log of the percentage of arable land and the log of an agricultural suitability index. A single continent dummy is used to represent the Americas, which is natural given the historical period examined. Regressions (3)–(6) do not employ the Oceania dummy due to a single observation for this continent in the corresponding regression samples, restricted by the availability of income per capita data. Robust standard error estimates are reported in parentheses.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Foreshadowing the qualitative robustness of the findings from previous sections, the logged indices of technology in the years 1000 CE and 1 CE are indeed highly correlated with the logged transition-timing variable. For instance, in the full cross-country samples employed by the population density regressions in Section IIIB, the logged Neolithic transition-timing variable possesses correlation coefficients of 0.73 and 0.62 with the logged indices of technology in the years 1000 CE and 1 CE, respectively. Similarly, in the income per capita data-restricted samples employed in Section IIIC, the corresponding correlation coefficients are 0.82 and 0.74.

Columns 1–2 reveal the full-sample regression results for population density in the years 1000 CE and 1 CE. Consistent with Malthusian predictions, the regressions indicate highly statistically significant positive relationships between technological sophistication and population density in the two time periods. To interpret the coefficients of interest, a 1 percent increase in the level of technological sophistication in the years 1000 CE and 1 CE corresponds to a rise in population density in the respective time periods by 4.32 and 4.22 percent, conditional on the productivity of land, geographical factors, and continental fixed effects.[33] In addition, columns 1–2 also indicate that the effects of the land-productivity channel on population density remain largely stable in comparison to previous estimates presented in Tables 3–4.

The results from replicating the 1000 CE and 1 CE analyses of Section IIIC, using the period-specific indices of technology as opposed to the timing of the Neolithic transition, are presented in columns 3–6. For each time period examined, the regressions for income per capita and population density reveal, exploiting identical variations in explanatory variables, that the estimated elasticity of population density with respect to the degree of technological sophistication is not only highly statistically significant, but at least an order of magnitude larger than the corresponding elasticity of income per capita. Indeed, the conditional correlation between technology and income per capita is not statistically different from zero at conventional levels of significance. A similar pattern also emerges for the estimated elasticities of population density and income per capita in each period with respect to the land-productivity channel. These findings therefore confirm the Malthusian prior that, in preindustrial times, variations in the level of technological advancement were ultimately manifested as variations in population density as opposed to variations in the standard of living across regions.

The remainder of the analysis in this section is concerned with establishing the causal effect of technology on population density in the years 1000 CE and 1 CE. Since the measures of technology employed by the preceding analysis are contemporaneous to population density in the two periods examined, the issue of endogeneity is perhaps more germane in this case than it was when examining the effect of the timing of the Neolithic Revolution on population density under the OLS estimator. In particular, the estimated coefficients associated with the period-specific technology indices in columns 1–2 of Table 7 may, in part, be capturing reverse causality, due to the potential scale effect of population on technological progress, as well as the latent influence of unobserved country-specific characteristics that are correlated with both technology and population density. To address these issues, the analysis to follow appeals to Diamond's (1997) argument, regarding the Neolithic transition to agriculture as a triggering event for subsequent technological progress, to exploit the exogenous component of cross-country variation in technology during the first millennium CE, as determined by the variation in the prehistoric biogeographical endowments that led to the differential timing of the Neolithic Revolution itself.[34]

The analysis proceeds by first establishing the causal effect of the Neolithic Revolution on subsequent technological progress. Given the high correlation between the prevalence of sedentary agricultural practices in Peregrine's (2003) dataset and the timing of the Neolithic transition, the current analysis exploits, for each period

[33]The partial regression lines associated with these coefficients appear in Figures W.5(a) and W.5(b) in the online Appendix.

[34]The potential issue of endogeneity arising from the latent influence of unobserved country fixed effects is also addressed by a first-difference estimation methodology employing data on population density and technological sophistication at two points in time. This strategy is pursued in Section IIIF below.

Table 8—The Causal Effect of the Neolithic Revolution on Technological Sophistication

Dependent variable is log non-agricultural technology in:	1000 CE			1 CE		
	OLS full sample (1)	OLS restricted sample (2)	IV restricted sample (3)	OLS full sample (4)	OLS restricted sample (5)	IV restricted sample (6)
Log years since Neolithic transition	0.115***	0.146***	0.279***	0.152***	0.174***	0.339***
	(0.024)	(0.030)	(0.073)	(0.027)	(0.029)	(0.074)
Log land productivity	−0.006	−0.012	−0.009	−0.024***	−0.027*	−0.023
	(0.008)	(0.015)	(0.014)	(0.008)	(0.016)	(0.019)
Log absolute latitude	0.012	0.000	0.005	0.039**	0.026	0.032
	(0.014)	(0.019)	(0.018)	(0.016)	(0.022)	(0.020)
Mean distance to nearest coast or river	0.008	0.117**	0.129**	0.007	0.050	0.066
	(0.033)	(0.053)	(0.051)	(0.035)	(0.084)	(0.078)
Percentage of land within 100 km of coast or river	0.024	0.080	0.112*	0.047	0.110	0.149**
	(0.038)	(0.052)	(0.058)	(0.048)	(0.070)	(0.076)
Continent dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	143	93	93	143	93	93
R^2	0.76	0.72	0.67	0.59	0.55	0.47
First-stage F-statistic	–	–	13.47	–	–	13.47
Overidentifying restrictions p-value	–	–	0.256	–	–	0.166

Notes: This table presents the causal effect of the timing of the Neolithic Revolution on the level of technology in non-agricultural sectors in the years 1000 CE and 1 CE, while controlling for land productivity, access to navigable waterways, absolute latitude, and unobserved continental fixed effects. Unlike the regular technology index, the index of non-agricultural technology for a given time period reflects the average degree of technological sophistication across only communications, transportation, and industrial sectors in that period. Log land productivity is the first principal component of the log of the percentage of arable land and the log of an agricultural suitability index. The IV regressions employ the numbers of prehistoric domesticable species of plants and animals as instruments for log transition timing. The statistic for the first-stage F-test of these instruments is significant at the 1 percent level. The p-values for the overidentifying restrictions tests correspond to Hansen's J statistic, distributed in both instances as χ^2 with one degree of freedom. A single continent dummy is used to represent the Americas, which is natural given the historical period examined. Regressions (2)–(3) and (5)–(6) do not employ the Oceania dummy due to a single observation for this continent in the IV data-restricted sample. Robust standard error estimates are reported in parentheses.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

examined, an alternative index of technological sophistication that is based only on the levels of technological advancement in communications, transportation, and industry, but otherwise identical in its underlying aggregation methodology to the index employed thus far. This permits a more transparent assessment of the argument that the Neolithic Revolution triggered a cumulative process of development, fueled by the emergence and propagation of a non-food producing class within agricultural societies that enabled sociocultural and technological advancements over and above subsistence activities.

Table 8 presents the results of regressions examining the impact of the timing of the Neolithic Revolution on the level of non-agricultural technological sophistication in the years 1000 CE and 1 CE, while controlling for land productivity, absolute latitude, access to waterways, and continental fixed effects. In line with priors, the regressions in columns 1 and 4 establish a highly statistically significant positive relationship between the timing of the Neolithic Revolution and the level of non-agricultural technological sophistication in each period, exploiting variation across the full sample of countries. To allow fair comparisons with the results from subsequent IV regressions, columns 2 and 5 repeat the preceding OLS analyses but on the subsample of countries for which data on the biogeographical instruments for the timing of the Neolithic Revolution are available. The results indicate that the OLS coefficients of interest from the preceding full-sample analyses remain robust to this change in the regression sample. Finally, columns 3 and 6 establish the causal effect of the Neolithic Revolution on the level of non-agricultural technological sophistication in the two time periods, employing the prehistoric availability of domesticable species of plants and animals as instruments for the timing of the Neolithic transition. Not surprisingly, as observed with earlier IV regressions, the causal impact of the Neolithic transition is, in each case, larger relative to its impact obtained under the OLS estimator, a pattern that is consistent with measurement error in the transition-timing variable and the resultant attenuation bias afflicting OLS coefficient estimates.

In light of the causal link between the timing of the Neolithic transition and the level of technological advancement in the first millennium CE, the analysis may now establish the causal impact of technology on population density in the two time periods examined. This is accomplished by exploiting exogenous variation in the level of technological advancement generated ultimately by differences in prehistoric biogeographical endowments that led to the differential timing of the transition to agriculture across countries. Table 9 reveals the results of this analysis where, as in Table 7, the measure of technology employed is the overall index that incorporates information on the prevalence of sedentary agriculture along with the level of advancement in non-agricultural technologies.

To facilitate comparisons of results obtained under the OLS and IV estimators, the full-sample OLS results from Table 7 for the years 1000 CE and 1 CE are again presented in columns 1 and 4 of Table 9, while columns 2 and 5 present the same regressions conducted on the IV-restricted subsample of countries. The causal effects of the level of technological advancement in the years 1000 CE and 1 CE, instrumented by the prehistoric availability of domesticable plant and animal species, on population density in the corresponding periods are revealed in columns 3 and 6. The estimated IV coefficients indicate a much larger causal impact of technology on population density, with a 1 percent increase in the level of technological sophistication in 1000 CE and 1 CE raising population density in the respective time periods by 14.53 and 10.80 percent, conditional on the productivity of land, absolute latitude, access to waterways, and continental fixed effects. Thus, in line with the predictions of the Malthusian theory, the results indicate that, during the agricultural stage of development, temporary gains due to improvements in the technological environment were indeed channeled into population growth, thereby leading more technologically advanced societies to sustain higher population densities.

E. *Robustness to Technology Diffusion and Geographical Factors*

This section establishes the robustness of the results for population density and income per capita in the year 1500 CE with respect to the spatial influence

TABLE 9—THE CAUSAL EFFECT OF TECHNOLOGICAL SOPHISTICATION ON POPULATION DENSITY

Dependent variable is log population density in:	1000 CE			1 CE		
	OLS full sample (1)	OLS restricted sample (2)	IV restricted sample (3)	OLS full sample (4)	OLS restricted sample (5)	IV restricted sample (6)
Log technology index in relevant period	4.315***	4.198***	14.530***	4.216***	3.947***	10.798***
	(0.850)	(1.164)	(4.437)	(0.745)	(0.983)	(2.857)
Log land productivity	0.449***	0.498***	0.572***	0.379***	0.350**	0.464**
	(0.056)	(0.139)	(0.148)	(0.082)	(0.172)	(0.182)
Log absolute latitude	−0.283**	−0.185	−0.209	−0.051	0.083	−0.052
	(0.120)	(0.151)	(0.209)	(0.127)	(0.170)	(0.214)
Mean distance to nearest coast or river	−0.638***	−0.363	−1.155*	−0.782***	−0.625	−0.616
	(0.188)	(0.426)	(0.640)	(0.198)	(0.434)	(0.834)
Percentage of land within 100 km of coast or river	0.385	0.442	0.153	0.237	0.146	−0.172
	(0.313)	(0.422)	(0.606)	(0.329)	(0.424)	(0.642)
Continent dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	140	92	92	129	83	83
R^2	0.61	0.55	0.13	0.62	0.58	0.32
First-stage F-statistic	–	–	12.52	–	–	12.00
Overidentifying restrictions p-value	–	–	0.941	–	–	0.160

Notes: This table presents the causal effect of direct measures of technological sophistication in the years 1000 CE and 1 CE, as determined by exogenous factors governing the timing of the Neolithic Revolution, on population density in the same time periods, while controlling for land productivity, access to navigable waterways, absolute latitude, and unobserved continental fixed effects. The technology index for a given time period reflects the average degree of technological sophistication across communications, transportation, industrial, and agricultural sectors in that period. The almost perfect collinearity between the degree of technological sophistication in the agricultural sector and the timing of the Neolithic transition does not permit the use of the latter as a covariate in these regressions. Log land productivity is the first principal component of the log of the percentage of arable land and the log of an agricultural suitability index. The IV regressions employ the numbers of prehistoric domesticable species of plants and animals as instruments for the log of the technology index in each of the two periods. In both cases, the statistic for the first-stage F-test of these instruments is significant at the 1 percent level. The p-values for the overidentifying restrictions tests correspond to Hansen's J statistic, distributed in both instances as χ^2 with one degree of freedom. A single continent dummy is used to represent the Americas, which is natural given the historical period examined. Regressions (2)–(3) and (5)–(6) do not employ the Oceania dummy due to a single observation for this continent in the IV data-restricted sample. Robust standard error estimates are reported in parentheses.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

of technological frontiers, as well as other geographical factors such as climate and small island and landlocked dummies, all of which may have had an effect on aggregate productivity either directly, by affecting the productivity of land, or indirectly, by affecting the prevalence of trade and technology diffusion. Specifically, the technology-diffusion hypothesis suggests that spatial proximity to societies at the world technology frontier confers a beneficial effect on development by facilitating the diffusion of new technologies from the frontier through trade as well as sociocultural and geopolitical influences. In particular, the diffusion channel implies that, ceteris paribus, the greater the geographical distance from the technological leaders in a given period, the lower the level of economic development amongst the followers in that period.

To account for the technology-diffusion channel, the current analysis employs, as a control variable, the great-circle distance from the capital city of a country to the closest of eight worldwide regional technological frontiers. These centers of technology diffusion are derived by Ashraf and Galor (2010), who employ historical urbanization estimates provided by Tertius Chandler (1987) and George Modelski (2003) to identify frontiers based on the size of urban populations. Specifically, for a given time period, their procedure selects, from each continent, the two largest cities in that period, belonging to distinct sociopolitical entities. Thus, the set of regional technological frontiers identified for the year 1500 CE comprises London and Paris in Europe, Fez and Cairo in Africa, Constantinople and Peking in Asia, and Tenochtitlan and Cuzco in the Americas.

Column 1 of Table 10 reveals the qualitative robustness of the full-sample regression results for population density in the year 1500 CE under controls for distance to the closest regional frontier as well as small island and landlocked dummies. To the extent that the gains from trade and technology diffusion are manifested primarily in terms of population size, as the Malthusian theory would predict, distance to the frontier has a highly statistically significant negative impact on population density. Nevertheless, the regression coefficients associated with the Neolithic transition-timing and land-productivity channels remain largely stable, albeit somewhat less so for the former, in comparison to their baseline estimates from column 4 in Table 2. Indeed, the lower magnitude of the coefficient associated with the transition-timing channel is attributable to the fact that several frontiers in the year 1500 CE, including Egypt, China, and Mexico, were also centers of diffusion of agricultural practices during the Neolithic Revolution and, as such, distance to the frontier in 1500 CE is partly capturing the effect of the differential timing of the Neolithic transition itself.

The regression in column 2 extends the robustness analysis of column 1 by adding controls for the percentage of land in temperate and tropical zones. The findings demonstrate that the effects of the Neolithic transition-timing, land-productivity, and spatial technology-diffusion channels on population density are indeed not spuriously driven by these additional climatological factors.

Columns 3–6 reveal the robustness of the results for income per capita as well as population density in the income per capita data-restricted sample, under controls for the technology-diffusion channel and additional geographical factors. In comparison to the relevant baseline regressions presented in columns 1 and 4 of Table 5, the coefficients associated with the transition-timing and land-productivity channels remain both qualitatively and quantitatively stable. In particular, the estimated elasticities of population density with respect to these channels are about an order of magnitude larger than the corresponding elasticities of income per capita regardless of the set of additional controls included in the specification.

With regard to the influence of technology diffusion, the qualitative pattern of the effects on population density versus income per capita is similar to those associated with the transition-timing and land-productivity channels. The finding that the negative elasticity of income per capita with respect to distance to the frontier is not only statistically insignificant but also at least an order of magnitude smaller than that of population density confirms Malthusian priors that the gains from trade and technology diffusion were primarily channeled into population growth

Table 10—Additional Robustness Checks

Dependent variable is:	Log population density in 1500 CE		Log income per capita in 1500 CE		Log population density in 1500 CE	
	OLS full sample (1)	OLS full sample (2)	OLS income sample (3)	OLS income sample (4)	OLS income sample (5)	OLS income sample (6)
Log years since Neolithic transition	0.828***	0.877***	0.117	0.103	1.498**	1.478**
	(0.208)	(0.214)	(0.221)	(0.214)	(0.546)	(0.556)
Log land productivity	0.559***	0.545***	0.036	0.047	0.596***	0.691***
	(0.048)	(0.063)	(0.032)	(0.037)	(0.123)	(0.122)
Log absolute latitude	−0.400***	−0.301**	−0.020	0.028	−0.354	0.668
	(0.108)	(0.129)	(0.110)	(0.247)	(0.392)	(0.783)
Mean distance to nearest coast or river	−0.403***	−0.388***	0.175	0.202	0.394	0.594
	(0.152)	(0.144)	(0.286)	(0.309)	(0.994)	(0.844)
Percentage of land within 100 km of coast or river	0.870***	0.837***	0.160	0.245	1.766***	2.491***
	(0.272)	(0.280)	(0.153)	(0.208)	(0.511)	(0.754)
Log distance to frontier	−0.186***	−0.191***	−0.005	−0.001	−0.130*	−0.108*
	(0.035)	(0.036)	(0.011)	(0.013)	(0.066)	(0.055)
Small island dummy	0.067	0.086	−0.118	−0.046	1.962**	2.720***
	(0.582)	(0.626)	(0.216)	(0.198)	(0.709)	(0.699)
Landlocked dummy	0.131	0.119	0.056	0.024	1.490***	1.269***
	(0.209)	(0.203)	(0.084)	(0.101)	(0.293)	(0.282)
Percentage of land in temperate zones		−0.196		−0.192		−1.624*
		(0.513)		(0.180)		(0.917)
Percentage of land in (sub)tropical zones		0.269		−0.025		1.153
		(0.307)		(0.308)		(1.288)
Continent dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	147	147	31	31	31	31
R^2	0.76	0.76	0.67	0.67	0.94	0.96

Notes: This table demonstrates that the relatively small effects of land productivity and the level of technological advancement, as proxied by the timing of the Neolithic Revolution, on income per capita in the year 1500 CE, but their significantly larger effects on population density in the same time period, remain robust under additional controls for technology diffusion and climatic factors. Log land productivity is the first principal component of the log of the percentage of arable land and the log of an agricultural suitability index. A single continent dummy is used to represent the Americas, which is natural given the historical period examined. Robust standard error estimates are reported in parentheses.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

rather than to improvements in living standards during preindustrial times.[35] While this finding may also be consistent with a non-Malthusian migration-driven theory of population movements against a spatial productivity gradient, the results uncovered by the first-difference estimation strategy pursued in the next section provide evidence in favor of the proposed Malthusian interpretation.

[35] Galor and Mountford (2008) reveal similar findings amongst non-OECD countries in the period spanning 1985–90, indicating that this phenomenon is more broadly associated with economies in the agricultural stage of development, even in the contemporary period.

F. *Robustness to Alternative Theories and Country Fixed Effects*

This section examines the robustness of the empirical findings to alternative theories and time-invariant country fixed effects. Specifically, the level regression results may be explained by the following non-Malthusian theory. In a world where labor is perfectly mobile, regions with higher aggregate productivity would experience labor inflows until regional wage rates were equalized, implying that, in levels, technology should be positively associated with population density but should not be correlated with income per capita across regions. Such a theory would also imply, however, that increases in the level of technology in any given region should generate increases in the standard of living in *all* regions. This runs contrary to the Malthusian prediction that increases in the level of technology in a given region should ultimately translate into increases in population density in that region, leaving income per capita constant at the subsistence level in all regions. Thus, examining the effect of a *change* in technology on *changes* in population density versus income per capita, as opposed to the impact of the *level* of technology on the *levels* of population density versus income per capita, constitutes a more discriminatory test of the Malthusian model.

Moreover, the level regressions in Table 7, indicating the significant positive relationship between the level of technology and population density but the absence of a systematic relationship with income per capita, could potentially reflect spurious correlations between technology and one or more unobserved time-invariant country fixed effects. By investigating the effect of changes on changes, however, one may "difference out" time-invariant country fixed effects, thereby ensuring that the coefficients of interest in the regression will not be afflicted by any such omitted variable bias. In addition, while the relationship between contemporaneous changes in technology and population density or income per capita could reflect reverse causality, this endogeneity issue may be alleviated somewhat by examining the impact of the *lagged* change in technology on changes in population density versus income per capita.

The current investigation thus examines the effect of the change in the level of technology between the years 1000 BCE and 1 CE on the change in population density, versus its effect on the change in income per capita, over the 1–1000 CE time horizon. In particular, the analysis compares the results from estimating the following empirical models:

$$\Delta \ln P_{i,t} = \mu_0 + \mu_1 \Delta \ln A_{i,t-1} + \phi_{i,t}, \tag{17}$$

$$\Delta \ln y_{i,t} = \nu_0 + \nu_1 \Delta \ln A_{i,t-1} + \psi_{i,t}, \tag{18}$$

where $\Delta \ln P_{i,t} \equiv \ln P_{i,t+1} - \ln P_{i,t}$ (i.e., the difference in log population density in country i between 1 CE and 1000 CE); $\Delta \ln y_{i,t} \equiv \ln y_{i,t+1} - \ln y_{i,t}$ (i.e., the difference in log income per capita of country i between 1 CE and 1000 CE); $\Delta \ln A_{i,t-1} \equiv \ln A_{i,t} - \ln A_{i,t-1}$ (i.e., the difference in log technology of country i between 1000 BCE and 1 CE); and, $\phi_{i,t}$ and $\psi_{i,t}$ are country-specific disturbance terms

TABLE 11—ROBUSTNESS TO ALTERNATIVE THEORIES AND TIME-INVARIANT COUNTRY FIXED EFFECTS

Dependent variable is diff. in:	Log population density between 1 CE and 1000 CE		Log income per capita between 1 CE and 1000 CE
	OLS full sample (1)	OLS income sample (2)	OLS income sample (3)
Diff. in log technology index between 1000 BCE and 1 CE	1.747*** (0.429)	3.133* (1.550)	0.073 (0.265)
Constant	0.451*** (0.053)	−0.026 (0.204)	−0.040 (0.064)
Observations	126	26	26
R^2	0.17	0.34	0.00

Notes: This table establishes that the change in the level of technological sophistication that occurred between the years 1000 BCE and 1 CE was primarily associated with a change in population density as opposed to a change in income per capita over the 1–1000 CE time horizon, and also reveals that there was no trend growth in income per capita during this period, thereby demonstrating robustness to time-invariant country fixed effects and dispelling an alternative migration-driven theory that is consistent with the level regression results. The technology index for a given time period reflects the average degree of technological sophistication across communications, transportation, industrial, and agricultural sectors in that period. The absence of controls from both regressions is justified by the removal of time-invariant country fixed effects through the application of the first-difference methodology. Tobust standard error estimates are reported in parentheses.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

for the changes in log population density and log income per capita. In addition, the intercept terms, μ_0 and ν_0, capture the average trend growth rates of population density and income per capita respectively over the 1–1000 CE time horizon. These models are the first-difference counterparts of (15) and (16), given that (i) $\ln A_{i,t-1}$ is used in lieu of $\ln T_i$, and (ii) the fixed effects of land productivity and the other geographical controls, including continental dummies, are time-invariant in those specifications.[36]

[36] In particular, equations (17) and (18) are obtained by applying the first-difference method to the following variants of equations (15) and (16):

$$\ln P_{i,t} = \gamma_0 + \mu_1 \ln A_{i,t-1} + \gamma_1 \ln X_i + \gamma_2' \mathbf{\Gamma}_i + \gamma_3' \mathbf{D}_i + \xi^P_{i,t},$$

$$\ln y_{i,t} = \lambda_0 + \nu_1 \ln A_{i,t-1} + \lambda_1 \ln X_i + \boldsymbol{\lambda}_2' \mathbf{\Gamma}_i + \boldsymbol{\lambda}_3' \mathbf{D}_i + \xi^y_{i,t},$$

with the respective error terms, $\xi^P_{i,t}$ and $\xi^y_{i,t}$, being modeled as:

$$\xi^P_{i,t} = \eta^P_i + \mu_0 t + \sigma^P_{i,t},$$

$$\xi^y_{i,t} = \eta^y_i + \nu_0 t + \sigma^y_{i,t},$$

where η^P_i and η^y_i are unobserved time-invariant country fixed effects on population density and income per capita in country i; μ_0 and ν_0 are global year fixed effects on population density and income per capita in year t; and, finally, $\sigma^P_{i,t}$ and $\sigma^y_{i,t}$ are country-year-specific disturbance terms for population density and income per capita. Thus, the error terms in equations (17) and (18) represent the changes over time in the aforementioned country-year-specific disturbance terms, i.e., $\phi_{i,t} \equiv \sigma^P_{i,t+1} - \sigma^P_{i,t}$ and $\psi_{i,t} \equiv \sigma^y_{i,t+1} - \sigma^y_{i,t}$. Strictly speaking, given that equations (15) and (16) allow for time-varying fixed effects, the actual first-difference counterparts of these equations, augmented with ln $A_{i,t-1}$ as an additional explanatory variable, would also have to control for transition timing, land productivity, and the

As discussed earlier, the alternative migration-driven theory predicts that an increase in technology in a given region will not differentially increase income per capita in that region due to the cross-regional equalization of wage rates, but will increase income per capita in all regions. In light of the specifications defined above, this theory would therefore imply that $\nu_1 = 0$ and $\nu_0 > 0$. According to the Malthusian theory, on the other hand, not only will the long-run level of income per capita remain unaffected in the region undergoing technological advancement, it will remain unaffected in all regions as well. The Malthusian theory thus implies that both $\nu_1 = 0$ and $\nu_0 = 0$.

Table 11 presents the results from estimating equations (17) and (18). As predicted by the Malthusian theory, the slope coefficients in columns 1 and 2 indicate that the change in the level of technology between the years 1000 BCE and 1 CE has a positive and statistically significant effect on the change in population density over the 1–1000 CE time horizon. In contrast, column 3 reveals that the corresponding effect on the change in income per capita over the time period 1–1000 CE is relatively marginal and not statistically significantly different from zero. Moreover, the intercept coefficient in column 3 suggests that the standard of living in 1000 CE was not significantly different from that in 1 CE, a finding that accords well with the Malthusian viewpoint. Overall, the results from the first-difference estimation strategy pursued in this section lend further credence to the Malthusian interpretation of the level regression results presented in earlier sections.

IV. Concluding Remarks

This paper examines the central hypothesis of the influential Malthusian theory, according to which improvements in the technological environment during the pre-industrial era had generated only temporary gains in income per capita, eventually leading to a larger, but not significantly richer, population. It exploits exogenous sources of cross-country variation in land productivity and technological levels to examine their hypothesized *differential* effects on population density versus income per capita during the time period 1–1500 CE.

Consistent with Malthusian predictions, the analysis uncovers statistically significant positive effects of land productivity and the technological level on population density in the years 1500 CE, 1000 CE and 1 CE. In contrast, the effects of land productivity and technology on income per capita in these periods are not significantly different from zero. Moreover, the estimated elasticities of income per capita with respect to these two channels are about an order of magnitude smaller than the corresponding elasticities of population density. Importantly, the qualitative results remain robust to controls for the confounding effects of a large number of geographical factors, including absolute latitude, access to waterways, distance to the technological frontier, and the share of land in tropical versus temperate climatic zones, which may have had an impact on aggregate productivity either directly, by affecting the productivity of land, or indirectly via the prevalence of trade and the

other baseline controls, including continental dummies. Results (not shown) from estimating these augmented first-difference specifications, however, are qualitatively similar to those obtained from estimating equations (17) and (18).

diffusion of technologies. Furthermore, the results are qualitatively unaffected under different measures of the level of technological advancement, including (i) the timing of the Neolithic Revolution and (ii) an index of technological sophistication. Finally, the study establishes that the results are not driven by unobserved time-invariant country fixed effects.

The analysis also dispels a non-Malthusian theory that may appear consistent with the level regression results. Specifically, in a world with perfect labor mobility, regions with higher aggregate productivity would have experienced labor inflows until regional wage rates were equalized, implying that technology should be positively associated with population density but should not be correlated with income per capita. However, labor inflows in response to technological improvements in a given region would result in higher income per capita in all regions, implying that changes in the level of technology should be positively associated with changes in the standard of living. On the contrary, using a first-difference estimation strategy with a lagged explanatory variable, the analysis demonstrates that, while changes in the level of technology between 1000 BCE and 1 CE were indeed associated with significant changes in population density over the 1–1000 CE time horizon, the level of income per capita across regions during this period was, in fact, largely unaffected, as suggested by the Malthusian theory.

In the course of the analysis, the paper generates three additional findings. First, in contrast to the positive relationship between absolute latitude and contemporary income per capita, population density in preindustrial times was on average higher at latitudinal bands closer to the equator. Thus, while proximity to the equator has been found to be detrimental in the industrial stage of development, it appears to have been beneficial during the agricultural stage. Second, the paper also establishes the importance of technological diffusion in the preindustrial world. To the extent that the gains from trade and technology diffusion are manifested primarily in terms of population size, as the Malthusian theory would predict, distance to the frontier has a highly statistically significant negative impact on population density. Finally, the analysis provides the first test of Diamond's (1997) influential hypothesis in the context of preindustrial societies, establishing that, indeed, an earlier onset of the Neolithic Revolution contributed to the level of technological sophistication and thus population density in the pre-modern world.

Interestingly, the epoch of Malthusian stagnation in income per capita masked a dynamism that may have ultimately brought about the phase transition that was associated with the take-off from the Malthusian regime. Although the growth of income per capita was minuscule over the Malthusian epoch, in the course of the Malthusian interaction between technology and population, technological progress intensified and world population significantly increased in size—a dynamism that was instrumental for the emergence of economies from the Malthusian trap.

Appendix

Table A.1—First-Stage Regressions

Second-stage dependent variable is:	Log of population density in 1500 CE	Log of population density in 1000 CE	Log of population density in 1 CE	Log of technology index in 1/1000 CE	Log of population density in 1000 CE	Log of population density in 1 CE
Endogenous variable is:	Log years since Neolithic transition				Log technology index in: 1000 CE	1 CE
	OLS (1)	OLS (2)	OLS (3)	OLS (4)	OLS (5)	OLS (6)
Excluded instruments:						
Domesticable plants	0.012**	0.013**	0.012**	0.012**	0.001	0.007***
	(0.005)	(0.005)	(0.006)	(0.005)	(0.001)	(0.002)
Domesticable animals	0.067**	0.064**	0.048*	0.063**	0.020***	−0.002
	(0.029)	(0.028)	(0.029)	(0.028)	(0.006)	(0.008)
Second-stage controls:						
Log land productivity	0.040	0.025	−0.011	0.023	0.002	−0.003
	(0.049)	(0.049)	(0.037)	(0.049)	(0.014)	(0.017)
Log absolute latitude	−0.127***	−0.130***	−0.083*	−0.120***	−0.015	−0.005
	(0.042)	(0.043)	(0.044)	(0.044)	(0.014)	(0.019)
Mean distance to nearest coast or river	0.127	0.103	0.094	0.079	0.112**	0.055
	(0.141)	(0.140)	(0.156)	(0.143)	(0.044)	(0.093)
Percentage of land within 100 km of coast or river	−0.165	−0.190	−0.227*	−0.171	0.044	0.061
	(0.137)	(0.136)	(0.136)	(0.137)	(0.036)	(0.063)
Continent dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	96	94	83	93	92	83
R^2	0.68	0.70	0.71	0.67	0.71	0.51
Partial R^2	0.27	0.28	0.25	0.26	0.17	0.16
F-statistic	14.65	15.10	10.85	13.47	12.52	12.00

Notes: This table collects the first-stage regression results for all IV regressions examined in the text. Specifically, regressions (1), (2), and (3) represent, respectively, the first stage of regression (6) in Tables 2, 3, and 4. Regression (4) corresponds to the first stage of both regressions (3) and (6) in Table 8. Finally, regressions (5) and (6) represent the first stage of regressions (3) and (6), respectively, in Table 9. Log land productivity is the first principal component of the log of the percentage of arable land and the log of an agricultural suitability index. The partial R^2 reported is for the excluded instruments only. The F-statistic is from the test of excluded instruments and is always significant at the 1 percent level; (iv) a single continent dummy is used to represent the Americas, which is natural given the historical period examined. The dummy for Oceania is not employed due to the presence of a single observation for this continent in the corresponding regression samples. Robust standard error estimates are reported in parentheses

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

REFERENCES

Aiyar, Shekhar, Carl-Johan Dalgaard, and Omer Moav. 2008. "Technological Progress and Regress in Pre-Industrial Times." *Journal of Economic Growth*, 13(2): 125–44.

Ashraf, Quamrul, and Oded Galor. 2007. "Cultural Assimilation, Cultural Diffusion and the Origin of the Wealth of Nations." Brown University Department of Economics Working Paper 2007-3.

Ashraf, Quamrul, and Oded Galor. 2010. "The 'Out of Africa' Hypothesis, Human Genetic Diversity, and Comparative Economic Development." Brown University Department of Economics Working Paper 2010-7.

Ashraf, Quamrul, and Oded Galor. 2011. "Dynamics and Stagnation in the Malthusian Epoch: Dataset: *American Economic Review*. http://www.aeaweb.org/articles.php?doi=10.1257/aer.101.5.2003.

Boserup, Ester. 1965. *The Conditions of Agricultural Growth: The Economics of Agrarian Change under Population Pressure*. London: George Allen & Unwin Ltd.

Chandler, Tertius. 1987. *Four Thousand Years of Urban Growth: An Historical Census*. Lewiston, NY: The Edwin Mellen Press.

Comin, Diego A., William Easterly, and Erick Gong. 2008. "Was the Wealth of Nations Determined in 1000 B.C.?" Harvard Business School Working Paper 09-052.

Conley, Timothy G. 1999. "GMM Estimation with Cross Sectional Dependence." *Journal of Econometrics*, 92(1): 1–45.

Crafts, Nicholas, and Terence C. Mills. 2009. "From Malthus to Solow: How Did the Malthusian Economy Really Evolve?" *Journal of Macroeconomics*, 31(1): 68–93.

Dalgaard, Carl-Johan, and Holger Strulik. 2010. "The Physiological Foundations of the Wealth of Nations." University of Copenhagen Department of Economics Discussion Paper 10-05.

Diamond, Jared. 1997. *Guns, Germs and Steel: The Fates of Human Societies*. New York, NY: W. W. Norton & Co.

Doepke, Matthias. 2004. "Accounting for Fertility Decline during the Transition to Growth." *Journal of Economic Growth*, 9(3): 347–83.

Galor, Oded. 2005. "From Stagnation to Growth: Unified Growth Theory." In *Handbook of Economic Growth, Vol IA*, ed. Philippe Aghion and Steven N. Durlauf, 171–293. Amsterdam: Elsevier North-Holland.

Galor, Oded. 2010. "The 2008 Lawrence R. Klein Lecture—Comparative Economic Development: Insights from Unified Growth Theory." *International Economic Review*, 51(1): 1–44.

Galor, Oded, and Omer Moav. 2002. "Natural Selection and the Origin of Economic Growth." *Quarterly Journal of Economics*, 117(4): 1133–91.

Galor, Oded, and Andrew Mountford. 2006. "Trade and the Great Divergence: The Family Connection." *American Economic Review*, 96(2): 299–303.

Galor, Oded, and Andrew Mountford. 2008. "Trading Population for Productivity: Theory and Evidence." *Review of Economic Studies*, 75(4): 1143–79.

Galor, Oded, and David N. Weil. 1999. "From Malthusian Stagnation to Modern Growth." *American Economic Review*, 89(2): 150–54.

Galor, Oded, and David N. Weil. 2000. "Population, Technology, and Growth: From Malthusian Stagnation to the Demographic Transition and Beyond." *American Economic Review*, 90(4): 806–28.

Hansen, Gary D., and Edward C. Prescott. 2002. "Malthus to Solow." *American Economic Review*, 92(4): 1205–17.

Kelly, Morgan, and Cormac Ó Gráda. 2010. "Living Standards and Mortality in England since the Middle Ages: The Poor Law versus the Positive Check." Unpublished.

Kremer, Michael. 1993. "Population Growth and Technological Change: One Million B.C. to 1990." *Quarterly Journal of Economics*, 108(3): 681–716.

Lagerlöf, Nils-Petter. 2003. "From Malthus to Modern Growth: Can Epidemics Explain the Three Regimes?" *International Economic Review*, 44(2): 755–77.

Lagerlöf, Nils-Petter. 2006. "The Galor-Weil Model Revisited: A Quantitative Exercise." *Review of Economic Dynamics*, 9(1): 116–42.

Lagerlöf, Nils-Petter. 2009. "Malthus in Sweden." Unpublished.

Livi-Bacci, Massimo. 2001. *A Concise History of World Population*. 3rd ed. Oxford: Blackwell Publishers Ltd.

Lucas, Robert E., Jr. 2002. "The Industrial Revolution: Past and Future." In *Lectures on Economic Growth*, ed. Robert E. Lucas, Jr., 109–88. Cambridge, MA: Harvard University Press.

Maddison, Angus. 2003. *The World Economy: Historical Statistics*. Paris: Organisation for Economic Co-operation and Development.

Maddison, Angus. 2008. "The West and the Rest in the World Economy: 1000–2030." *World Economics*, 9(4): 75–99.

Malthus, Thomas R. 1798. *An Essay on the Principle of Population*. Reprint, ed. Geoffrey Gilbert, 1999. Oxford: Oxford University Press.

McEvedy, Colin, and Richard Jones. 1978. *Atlas of World Population History*. New York: Penguin Books.

Michalopoulos, Stelios. 2008. "The Origins of Ethnolinguistic Diversity: Theory and Evidence." Tufts University Department of Economics Discussion Paper 0725.

Modelski, George. 2003. *World Cities: −3000 to 2000*. Washington, DC: FAROS 2000.

Olsson, Ola, and Douglas A. Hibbs, Jr. 2005. "Biogeography and Long-Run Economic Development." *European Economic Review*, 49(4): 909–38.

O'Rourke, Kevin H., Ahmed S. Rahman, and Alan M. Taylor. 2008. "Luddites and the Demographic Transition." National Bureau of Economic Research Working Paper 14484.

Peregrine, Peter N. 2003. "Atlas of Cultural Evolution." *World Cultures: Journal of Comparative and Cross-Cultural Research*, 14(1): 1–75.

Putterman, Louis. 2008. "Agriculture, Diffusion and Development: Ripple Effects of the Neolithic Revolution." *Economica*, 75(300): 729–48.

Ramankutty, Navin, Jonathan A. Foley, John Norman, and Kevin McSweeney. 2002. "The Global Distribution of Cultivable Lands: Current Patterns and Sensitivity to Possible Climate Change." *Global Ecology and Biogeography*, 11(5): 377–92.

Strulik, Holger, and Jacob Weisdorf. 2008. "Population, Food, and Knowledge: A Simple Unified Growth Theory." *Journal of Economic Growth*, 13(3): 195–216.

Voigtländer, Nico, and Hans-Joachim Voth. 2006. "Why England? Demographic Factors, Structural Change and Physical Capital Accumulation during the Industrial Revolution." *Journal of Economic Growth*, 11(4): 319–61.

Voigtländer, Nico, and Hans-Joachim Voth. 2009. "Malthusian Dynamism and the Rise of Europe: Make War, Not Love." *American Economic Review*, 99(2): 248–54.

Weisdorf, Jacob L. 2005. "From Foraging to Farming: Explaining the Neolithic Revolution." *Journal of Economic Surveys*, 19(4): 561–86.

Weisdorf, Jacob. 2009. "Why Did the First Farmers Toil? Human Metabolism and the Origins of Agriculture." *European Review of Economic History*, 13(2): 157–72.